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PURE MATHEMATICS WITH STATISTICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH STATISTICS

CHAPTER 10

ELEMENTARY PROBABILITY

SUGGESTED TIME: 15 PERIODS

IN THIS CHAPTER

This chapter introduces the language and laws of probability: experiments, sample spaces, events, classical and empirical probability, the addition and multiplication laws, conditional probability, tree diagrams, counting techniques and Bayes' theorem. It provides the essential foundation for the statistics and probability questions of the GCE Advanced Level Paper in Pure Mathematics with Statistics.

- By the end of this chapter I can define probability and use correctly the basic vocabulary: experiment, outcome, sample space, event.
- By the end of this chapter I can list the sample space of simple and compound experiments and identify types of events (simple, compound, complementary, mutually exclusive, exhaustive, independent).
- By the end of this chapter I can compute classical (theoretical) probabilities using equally likely outcomes.
- By the end of this chapter I can estimate probabilities empirically from relative frequencies and relate them to theoretical probability.
- By the end of this chapter I can apply the addition law for mutually exclusive and for non-mutually exclusive events.
- By the end of this chapter I can apply the multiplication law and compute conditional probabilities, testing events for independence.

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COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Probability, Experiments and Sample Spaces
2. Types of Events and Classical Probability
3. Empirical Probability and the Addition Laws
4. Multiplication Law, Conditional Probability and Independence
5. Tree Diagrams and Selection With or Without Replacement
6. Counting Techniques in Probability
7. Total Probability and Bayes' Theorem
8. Integration activity
9. Methods and worked examples
10. Exercises
11. Solutions to the exercises
12. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can define probability and use correctly the basic vocabulary: experiment, outcome, sample space, event.
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- By the end of this chapter I can apply the addition law for mutually exclusive and for non-mutually exclusive events.
- By the end of this chapter I can apply the multiplication law and compute conditional probabilities, testing events for independence.
- By the end of this chapter I can construct and use tree diagrams, including selections with and without replacement.
- By the end of this chapter I can use permutations and combinations to count outcomes in probability problems.
- By the end of this chapter I can state and apply Bayes' theorem to reverse conditional probabilities in real-life contexts.

PREREQUISITES

- Set language and operations: union, intersection, complement, Venn diagrams (Lower Sixth).
- Fractions, decimals, percentages and their simplification.
- Factorial notation, permutations (nPr) and combinations (nCr) from the chapter on counting techniques.
- Basic algebraic manipulation: solving simple linear equations and working with fractions of fractions.
- Interpreting simple statistical tables and relative frequencies (Lower Sixth statistics).

1 PROBABILITY, EXPERIMENTS AND SAMPLE SPACES

Every rainy season in Douala, commuters at Ndokoti junction gamble on whether the sky will open before they reach Bonabéri; every Saturday, punters at a Yaoundé betting kiosk stake money on football outcomes they cannot control. Behind each of these everyday decisions lies the same question: how likely is an uncertain event? Probability gives a precise numerical answer between 0 and 1, and it powers insurance premiums, medical testing, election forecasts and the National Lottery. In this chapter you will learn to measure chance rigorously, from tossing a coin to applying Bayes' theorem to a medical diagnosis.

To measure chance, we must first agree on the language we use to describe uncertain situations. That is the goal of this first section: we define what a **random experiment** is, we list all its possible results in a **sample space**, and we learn to describe **events** as sets of results. Everything else in the chapter is built on these foundations.

1.1 Deterministic and Random Experiments

In science and in daily life, we perform experiments: actions whose results we observe. Some experiments are completely predictable.

EXPERIMENT **Two kinds of experiments**

Experiment 1. Heat pure water at sea level. It boils at 100 °C, every single time. The result is fixed by the laws of physics.
Experiment 2. Toss a 25 FCFA coin and note which face is up. Sometimes it is Head, sometimes Tail. No matter how carefully you repeat the toss under identical conditions, you cannot predict the result of the next toss.

Experiment 1 is **deterministic**: the same conditions always produce the same result. Experiment 2 is **random**: the result cannot be predicted in advance, even though we know all the possible results.

DEFINITION **Random experiment, trial, outcome**

A **random experiment** is an action or process whose result cannot be predicted with certainty before it is performed, but whose set of all possible results is known.
Each performance of the experiment is called a **trial**, and each possible result of a trial is called an **outcome**.

EXEMPLE **Classifying experiments**

Classify each experiment as deterministic or random.

1. Dropping a stone from the top of a building in Buea and measuring whether it falls. De-

terministic — it always falls.

2. Throwing a fair die and recording the number on top. Random — any of 1, 2, 3, 4, 5, 6 may appear.
3. Drawing one card from a well-shuffled pack of 52 cards. Random.
4. Counting the number of days in the month of February 2024. Deterministic — the answer, 29, is fixed.

1.2 The Sample Space

Before we can measure how likely something is, we must know everything that can happen. This complete list is called the sample space.

DEFINITION **Sample space**

The **sample space** of a random experiment, denoted S (or Ω), is the set of all possible outcomes of the experiment.

EXEMPLE **Sample spaces of one-stage experiments**

1. Tossing one coin: $S = \{H, T\}$, where H = Head and T = Tail.
2. Throwing one die: $S = \{1, 2, 3, 4, 5, 6\}$.
3. Drawing one card and noting its suit: $S = \{\text{Hearts, Diamonds, Clubs, Spades}\}$.
4. Spinning a spinner with four coloured sectors: $S = \{\text{Red, Blue, Green, Yellow}\}$.

1.3 Sample Spaces of Two-Stage Experiments

Many experiments have two stages: two coins are tossed, two dice are thrown, or a coin is tossed and then a die is thrown. Listing the sample space carelessly almost always leads to missing or duplicated outcomes. We therefore use **systematic enumeration**: ordered pairs, lists, or two-way tables (grids).

METHOD **Systematic listing of a sample space**

1. Identify the stages of the experiment and the possible results of each stage.
2. Represent each outcome as an ordered pair (stage 1 result, stage 2 result).

3. For two-stage experiments, draw a two-way table: rows for the first stage, columns for the second stage; each cell is one outcome.
4. Count the outcomes: if stage 1 has m results and stage 2 has n results, the sample space has $m \times n$ outcomes.

EXEMPLE **Two coins**

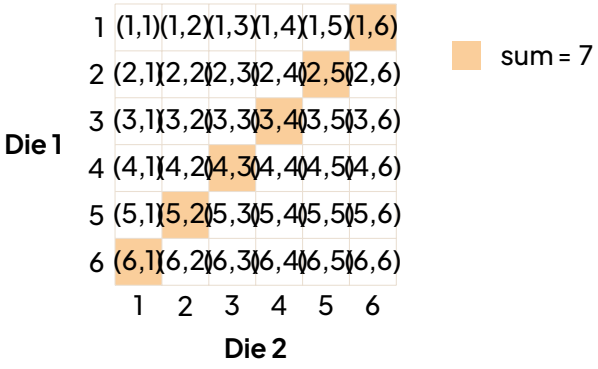
Tossing two coins: stage 1 gives H or T, stage 2 gives H or T. The sample space is

$$S = \{(H, H), (H, T), (T, H), (T, T)\},$$

which has $2 \times 2 = 4$ outcomes. Note that (H, T) and (T, H) are different outcomes: in the first, coin 1 shows Head; in the second, coin 2 shows Head.

EXEMPLE **Two dice**

Throwing two dice (one red, one blue): each die has 6 faces, so the sample space has $6 \times 6 = 36$ outcomes, each written (red, blue). The grid below displays all 36 outcomes and highlights the event "the sum of the two numbers is 7".



The 36 outcomes of throwing two dice. The shaded diagonal cells are the six outcomes whose coordinates sum to 7.

EXEMPLE **A coin, then a die**

Toss a coin, then throw a die. The sample space is

$$S = \{(H,1),(H,2),(H,3),(H,4),(H,5),(H,6),(T,1),(T,2),(T,3),(T,4),(T,5),(T,6)\},$$

with $2 \times 6 = 12$ outcomes.

1.4 Events and Types of Events

Usually we are not interested in one single outcome but in a collection of outcomes: "an even number on the die", "at least one Head", "a sum of 7". Such collections are called events.

DEFINITION **Event, simple event, compound event**

An **event** is any subset of the sample space S. We say an event A occurs when the outcome of the trial belongs to A.

- A **simple** (or **elementary**) **event** contains exactly one outcome.
- A **compound event** contains two or more outcomes.
- The whole sample space S is the **certain event**: it always occurs.
- The empty set $\varnothing$ is the **impossible event**: it never occurs.

EXEMPLE **Events when throwing one die**

With $S = \{1,2,3,4,5,6\}$:

- $A = \{5\}$ is a simple event ("score 5").
- $B = \{2,4,6\}$ is a compound event ("even score").
- $C = \{1,2,3,4,5,6\} = S$ is the certain event ("score less than 7").
- $D = \varnothing$ is the impossible event ("score 8").

WARNING **An outcome is not an event**

Do not confuse an outcome with an event. The outcome is a single result, e.g. 3. An event is a set of outcomes, e.g. $\{3\}$ or $\{1,3,5\}$. Writing "the event 3" is incorrect; write "the event $\{3\}$". This distinction becomes essential when we compute probabilities: we always compute the probability of a set.

1.5 Types of Events: Mutually Exclusive and Exhaustive Events

Beyond simple and compound, events are classified by how they relate to each other.

DEFINITION **Mutually exclusive events**

Two events A and B are **mutually exclusive** (or **disjoint**) if they cannot occur simultaneously in the same trial. In set notation:

$$A \cap B = \varnothing.$$

DEFINITION **Exhaustive events**

A set of events $A_1, A_2, \ldots, A_n$ is **exhaustive** if at least one of them must occur in every trial. In set notation:

$$A_1 \cup A_2 \cup \cdots \cup A_n = S.$$

If the events are both mutually exclusive and exhaustive, they form a **partition** of the sample space.

EXEMPLE **Mutually exclusive and exhaustive events**

Throwing one die: $S = \{1, 2, 3, 4, 5, 6\}$.

- $A = \{1, 2, 3\}$ (score ≤ 3) and $B = \{4, 5, 6\}$ (score ≥ 4) are mutually exclusive ($A \cap B = \varnothing$) and exhaustive ($A \cup B = S$). They form a partition.
- $C = \{2, 4, 6\}$ (even) and $D = \{1, 3, 5\}$ (odd) are also mutually exclusive and exhaustive.
- $E = \{1, 2\}$ and $F = \{2, 3\}$ are not mutually exclusive ($E \cap F = \{2\} \neq \varnothing$).
- $G = \{1, 2\}$ and $H = \{4, 5\}$ are mutually exclusive but not exhaustive ($G \cup H = \{1, 2, 4, 5\} \neq S$).

PROPERTY **Probability of mutually exclusive events (preview)**

If A and B are mutually exclusive, then $P(A \cap B) = 0$. The full Addition Law is developed in Section 3.

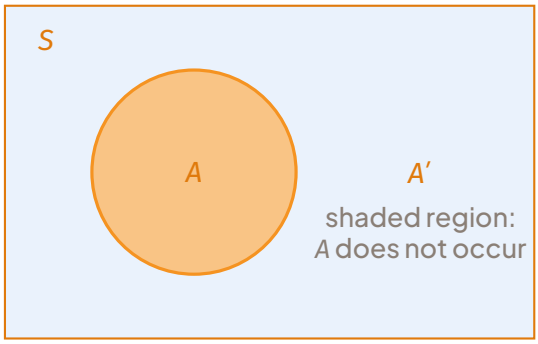
1.6 The Complement of an Event

DEFINITION **Complement**

The **complement** of an event A, written A' (or $\bar{A}$), is the event that A does not occur. It contains every outcome of S that is not in A:

$$A' = \{\text{outcomes in } S \text{ but not in } A\}.$$

For example, throwing one die with $A = \{2, 4, 6\}$ (even score) gives $A' = \{1, 3, 5\}$ (odd score). Since every trial produces exactly one outcome, either A occurs or A' occurs, never both and never neither. Consequently, A and A' are always mutually exclusive and exhaustive: $A \cap A' = \varnothing$ and $A \cup A' = S$.



The sample space S (rectangle) containing an event A (circle). The region outside the circle but inside the rectangle is the complement A'.

1.7 The Probability Scale (Intuitive Introduction)

We now attach a number to each event. Intuitively, probability measures how likely an event is, on a scale from 0 to 1:

- $P = 0$: the event is **impossible** (it can never happen);
- $P = 1$: the event is **certain** (it must happen);
- values in between measure increasing likelihood: $P = 0.5$ means "as likely as not".

PROPERTY **Fundamental properties of probability**

For any event A in a sample space S:

$$0 \le P(A) \le 1, \qquad P(S) = 1, \qquad P(\varnothing) = 0, \qquad P(A) + P(A') = 1.$$

(These properties follow directly from the definition of probability as a proportion of outcomes; they are not examinable as proofs but must be known and used fluently.)

WARNING **Probabilities outside** $[0, 1]$

A probability can never be negative or greater than 1. If your calculation gives $P(A) = 1.4$ or $P(A) = -0.2$, you have made an error — go back and check. Also, $P(A) = 0$ means the event is impossible; it does not mean "very unlikely".

1.8 Equally Likely Outcomes and Fairness

To assign probabilities we need one more ingredient: the idea that some outcomes are equally likely.

DEFINITION **Equally likely outcomes**

Outcomes of a random experiment are **equally likely** when no outcome is more likely to occur than any other. This is the case for a fair coin, an unbiased die, or a well-shuffled pack of cards.

When a fair coin is tossed, $P(H) = P(T) = 0.5$. When an unbiased die is thrown, each face has probability $\frac{1}{6}$. The assumption of fairness is a modelling choice: a real coin may be slightly imperfect, but the fair model is an excellent approximation — and the GCE will always tell you when a coin or die is biased.

WORKED EXAMPLE 1 **Sample space, events and complements**

Two fair coins are tossed.

1. Write down the sample space S.
2. Write as sets the events: A = "exactly one Head", B = "at least one Head", C = "both coins show the same face".
3. Write down B' and verify that $B \cap B' = \varnothing$ and $B \cup B' = S$.
4. Which of A, B, C are simple events and which are compound?
5. Are A and C mutually exclusive? Are they exhaustive?

SOLUTION

Solution.

1. $S = \{(H,H), (H,T), (T,H), (T,T)\}$.
2. $A = \{(H,T), (T,H)\}$; $B = \{(H,H), (H,T), (T,H)\}$; $C = \{(H,H), (T,T)\}$.
3. B' = "no Head" = $\{(T,T)\}$. Then $B \cap B' = \varnothing$ (no outcome is in both) and $B \cup B' =$

$\{(H,H),(H,T),(T,H),(T,T)\} = S$, as required for complementary events.

4. A, B and C each contain more than one outcome, so all three are compound events. The simple events here are $\{(H,H)\}$, $\{(H,T)\}$, $\{(T,H)\}$ and $\{(T,T)\}$.

5. $A \cap C = \varnothing$ (no outcome has exactly one Head and both coins the same), so A and C are mutually exclusive. $A \cup C = \{(H,T),(T,H),(H,H),(T,T)\} = S$, so they are also exhaustive. Hence A and C form a partition of S.

EXERCISE 1 **Building sample spaces and identifying events**

1. A fair die is thrown and a fair coin is tossed. List the sample space and state the number of outcomes.

2. A bag contains 3 balls: one red (R), one green (G), one blue (B). Two balls are drawn, one after the other, the first being replaced before the second draw. List the sample space.

3. For the experiment in (2), write as sets the events E = "both balls have the same colour" and F = "the red ball is drawn at least once". Then write E' and F'.

4. In the experiment of (1), let G = "the die shows an even number" and H = "the coin shows Tail". Are G and H mutually exclusive? Are they exhaustive?

SOLUTION **Exercise 1**

1. $S = \{(1,H),(2,H),(3,H),(4,H),(5,H),(6,H),(1,T),(2,T),(3,T),(4,T),(5,T),(6,T)\}$; there are $6 \times 2 = 12$ outcomes.

2. $S = \{(R,R),(R,G),(R,B),(G,R),(G,G),(G,B),(B,R),(B,G),(B,B)\}$; $3 \times 3 = 9$ outcomes.

3. $E = \{(R,R),(G,G),(B,B)\}$; $F = \{(R,R),(R,G),(R,B),(G,R),(B,R)\}$; $E' = \{(R,G),(R,B),(G,R),(G,B),(B,R),(B,G)\}$ (the two balls have different colours); $F' = \{(G,G),(G,B),(B,G),(B,B)\}$ (the red ball is not drawn).

4. $G = \{(2,H),(4,H),(6,H),(2,T),(4,T),(6,T)\}$; $H = \{(1,T),(2,T),(3,T),(4,T),(5,T),(6,T)\}$. $G \cap H = \{(2,T),(4,T),(6,T)\} \neq \varnothing$, so not mutually exclusive. $G \cup H = \{(1,T),(2,H),(2,T),(3,T),(4,H),(4,T),(5,T),(6,H),(6,T)\} \neq S$ (missing $(1,H),(3,H),(5,H)$), so not exhaustive.

KEY POINTS **Key points of this section**

- A **random experiment** has an unpredictable result; a **trial** is one performance; an **outcome** is one possible result.
- The **sample space** S is the set of all outcomes. For two-stage experiments, use ordered pairs and two-way tables: m results then n results give $m \times n$ outcomes.
- An **event** is a subset of S: simple (one outcome) or compound (several). S is certain, $\varnothing$ is impossible.
- **Mutually exclusive** events cannot occur together ($A \cap B = \varnothing$). **Exhaustive** events cover all possibilities ($A \cup B = S$). A partition is both.
- The **complement** A' is "A does not occur", with $A \cap A' = \varnothing$, $A \cup A' = S$, and later $P(A) + P(A') = 1$.
- For every event: $0 \le P(A) \le 1$, $P(S) = 1$, $P(\varnothing) = 0$.
- **Equally likely** outcomes (fair coin, unbiased die) are the basis of classical probability, developed in the next section.

2 TYPES OF EVENTS AND CLASSICAL PROBABILITY

In the previous section we introduced the language of probability: experiments, outcomes, sample spaces and events. We now move to the heart of the matter — how to assign a numerical value to the likelihood of an event. The oldest and most intuitive approach is **classical (theoretical) probability**, which applies when all outcomes of an experiment are **equally likely**. We shall also classify events according to how they relate to each other (mutually exclusive, exhaustive, complementary) and see how these relationships simplify calculations.

2.1 Classical Probability — Equally Likely Outcomes

EXPERIMENT **Observing symmetry in games of chance**

Take a fair six-sided die. There is no physical reason why the face 3 should appear more often than the face 5: the cube is symmetric, the mass is evenly distributed, and each face has the same shape and size. Common sense therefore tells us that each of the six faces has the same chance, namely $\frac{1}{6}$. The same reasoning applies to a fair coin (two symmetric faces) or to a well-shuffled pack of cards (no card is favoured). This symmetry principle is the foundation of classical probability.

DEFINITION **Classical Probability**

If a random experiment has a finite sample space S in which all $n(S)$ outcomes are equally likely, and A is an event consisting of $n(A)$ outcomes, then the probability of A is

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}.$$

PROPERTY **Elementary properties of probability**

For any experiment with finite sample space S and equally likely outcomes:

- $0 \le P(A) \le 1$ for every event A;
- $P(S) = 1$ (the certain event) and $P(\varnothing) = 0$ (the impossible event);
- $P(A) + P(A') = 1$, where A' is the complement of A.

These follow immediately from the definition, since $0 \le n(A) \le n(S)$, $n(S)/n(S) = 1$, $n(\varnothing) = 0$ and $n(A) + n(A') = n(S)$.

WARNING **Equally likely outcomes only!**

The formula $P(A) = n(A)/n(S)$ is valid only when every outcome in S has the same chance of occurring. Do not apply it to biased coins, loaded dice, or situations where outcomes have different probabilities (e.g. the weather tomorrow). In such cases we must use empirical probability (next section) or another model. A frequent GCE error is to count outcomes that are not equally likely: for example, when two coins are tossed, the three events "two heads", "one head and one tail", "two tails" are not equally likely — the correct sample space is $\{HH, HT, TH, TT\}$ with four equally likely outcomes.

EXEMPLE **Rolling a fair die**

A fair six-sided die is rolled once. The sample space is $S = \{1, 2, 3, 4, 5, 6\}$ with $n(S) = 6$ equally likely outcomes.

- Event A: "an even number appears" $\Rightarrow A = \{2, 4, 6\}$, $n(A) = 3$. $P(A) = \frac{3}{6} = \frac{1}{2}$.
- Event B: "a number greater than 4" $\Rightarrow B = \{5, 6\}$, $n(B) = 2$. $P(B) = \frac{2}{6} = \frac{1}{3}$.
- Event C: "a 7 appears" $\Rightarrow C = \varnothing$, $n(C) = 0$. $P(C) = 0$ (impossible event).
- Event D: "a number less than 7" $\Rightarrow D = S$, $n(D) = 6$. $P(D) = 1$ (certain event).

Standard 52-Card Pack

	A	2	3	4	5	6	7	8	9	10	J	Q	K
S	A♠	2♠	3♠	4♠	5♠	6♠	7♠	8♠	9♠	10♠	J♠	Q♠	K♠
H	A♥	2♥	3♥	4♥	5♥	6♥	7♥	8♥	9♥	10♥	J♥	Q♥	K♥
D	A♦	2♦	3♦	4♦	5♦	6♦	7♦	8♦	9♦	10♦	J♦	Q♦	K♦
C	A♣	2♣	3♣	4♣	5♣	6♣	7♣	8♣	9♣	10♣	J♣	Q♣	K♣

Structure of a standard 52-card deck: 4 suits × 13 ranks. S = Spades, C = Clubs (black cards); H = Hearts, D = Diamonds (red cards). A = Ace, J = Jack, Q = Queen, K = King.

EXEMPLE **Drawing a card from a well-shuffled pack**

A standard pack has 52 cards. One card is drawn at random.

- $P(\text{Ace}) = \frac{4}{52} = \frac{1}{13}$.
- $P(\text{Heart}) = \frac{13}{52} = \frac{1}{4}$.
- $P(\text{Ace of Hearts}) = \frac{1}{52}$.
- $P(\text{Face card: J, Q or K}) = \frac{12}{52} = \frac{3}{13}$.
- $P(\text{Red card}) = \frac{26}{52} = \frac{1}{2}$.

EXEMPLE **Balls in a bag**

A bag contains 5 red, 3 blue and 2 green balls (total 10). One ball is drawn at random.

$$P(\text{Red}) = \frac{5}{10} = \frac{1}{2}, \quad P(\text{Blue}) = \frac{3}{10}, \quad P(\text{Green}) = \frac{2}{10} = \frac{1}{5}, \quad P(\text{Not red}) = \frac{5}{10} = \frac{1}{2}.$$

Note that $P(\text{Red}) + P(\text{Blue}) + P(\text{Green}) = \frac{1}{2} + \frac{3}{10} + \frac{1}{5} = 1$: the three events are mutually exclusive and exhaustive.

EXEMPLE **Lottery-type draw**

In a lottery, 6 numbers are drawn from $\{1, 2, \ldots, 49\}$ without replacement, and the order does not matter. The total number of equally likely combinations is $\binom{49}{6} = 13\,983\,816$. If

you buy one ticket with a specific combination, your chance of winning the jackpot is

$$P(\text{jackpot}) = \frac{1}{\binom{49}{6}} = \frac{1}{13\,983\,816} \approx 7.15 \times 10^{-8}.$$

This tiny number explains why jackpot winners are so rare!

2.2 Relationships Between Events

Events are subsets of the sample space S. We classify them by how they overlap.

DEFINITION **Mutually Exclusive (Disjoint) Events**

Two events A and B are mutually exclusive if they cannot occur simultaneously, i.e. $A \cap B = \varnothing$. On a Venn diagram their regions do not overlap. For example, when a die is rolled once, "obtaining a 2" and "obtaining a 5" are mutually exclusive.

DEFINITION **Exhaustive Events**

A collection of events $A_1, A_2, \ldots, A_k$ is exhaustive if their union is the whole sample space: $A_1 \cup A_2 \cup \cdots \cup A_k = S$. In other words, at least one of them must occur. If they are also pairwise mutually exclusive, they form a partition of S, and then

$$P(A_1) + P(A_2) + \cdots + P(A_k) = 1.$$

DEFINITION **Complementary Event**

For any event A, its complement A' (also written $\overline{A}$ or A^c) is the event "A does not occur". A and A' are mutually exclusive and exhaustive: $A \cap A' = \varnothing$ and $A \cup A' = S$. Hence $P(A) + P(A') = 1$, so

$$P(A') = 1 - P(A).$$

DEFINITION **Independent Events (informal)**

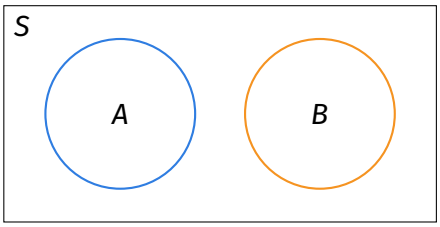
Two events A and B are independent if the occurrence of one does not affect the probability of the other. The formal definition, $P(A \cap B) = P(A)P(B)$, will be given in a later section. For now, think of "drawing a card, replacing it, shuffling, and drawing again" — the two draws are independent.

WARNING **Mutually exclusive ≠ Independent**

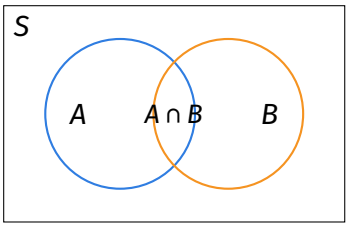
This is a classic trap!

- Mutually exclusive: A and B cannot happen together $\Rightarrow$ $P(A \cap B) = 0$. If $P(A) > 0$ and $P(B) > 0$, they are in fact dependent: knowing that A occurred tells you that B certainly did not.
- Independent: $P(A \cap B) = P(A)P(B)$. Two independent events with non-zero probabilities can happen together.

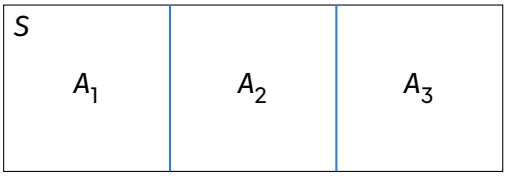
Never write $P(A \cap B) = P(A)P(B)$ for mutually exclusive events, and never write $P(A \cap B) = 0$ for independent events.



(a) Mutually exclusive: $A \cap B = \varnothing$



(b) Not mutually exclusive: $A \cap B \neq \varnothing$



(c) Exhaustive partition: $A_1 \cup A_2 \cup A_3 = S$, pairwise disjoint

Venn diagrams: (a) mutually exclusive events; (b) overlapping events; (c) three events forming an exhaustive partition of S.

2.3 Using the Complement — "At Least One" Strategy

METHOD **Complement Rule for "At Least One"**

When a problem asks for the probability of "at least one" success in several trials, it is often much easier to compute the probability of none (the complement) and subtract from 1.

1. Identify the event "no success" (all trials fail).

2. Compute $P(\text{no success})$ — usually a product if the trials are independent.
3. Then $P(\text{at least one success}) = 1 - P(\text{no success})$.

EXEMPLE **At least one six in two rolls of a fair die**

Roll a fair die twice. Find $P(\text{at least one 6})$.

$$P(\text{no 6 in one roll}) = \frac{5}{6}.$$
$$P(\text{no 6 in two rolls}) = \left(\frac{5}{6}\right)^2 = \frac{25}{36}.$$
$$P(\text{at least one 6}) = 1 - \frac{25}{36} = \frac{11}{36}.$$

EXEMPLE **At least one ace in a 5-card hand**

A hand of 5 cards is dealt from a standard 52-card pack. Find $P(\text{at least one ace})$.

$$P(\text{no ace}) = \frac{\binom{48}{5}}{\binom{52}{5}} = \frac{1\,712\,304}{2\,598\,960} \approx 0.6588.$$

$$P(\text{at least one ace}) = 1 - 0.6588 = 0.3412.$$

Direct counting would require four separate cases (exactly 1, 2, 3 or 4 aces); the complement gives the answer in one line.

2.4 Odds in Favour and Odds Against (GCE Speed Tip)

DEFINITION **Odds**

If $P(A) = p$ (with $0 < p < 1$), then

- Odds in favour of $A = p : (1 - p)$, usually simplified to a ratio $a : b$ of integers;
- Odds against $A = (1 - p) : p$.

Conversely, if the odds in favour of A are $a : b$, then $P(A) = \frac{a}{a+b}$.

EXEMPLE **Odds examples**

- Fair die: $P(6) = \frac{1}{6}$. Odds in favour of a 6 = 1 : 5. Odds against = 5 : 1.
- Drawing a heart from a pack: $P(\text{heart}) = \frac{1}{4}$. Odds in favour = 1 : 3.
- If the odds against an event are 7 : 2, then $P(\text{event}) = \frac{2}{7+2} = \frac{2}{9}$.

2.5 Worked Examination-Style Examples

WORKED EXAMPLE 2 **Two dice — sum and product**

Two fair six-sided dice are rolled. Find the probability that:

1. the sum of the scores is 8;
2. the product of the scores is 12;
3. the sum is 8 or the product is 12.

SOLUTION

Solution. The sample space has $n(S) = 6 \times 6 = 36$ equally likely ordered pairs (d_1, d_2).
1. Sum = 8: favourable pairs $(2,6), (3,5), (4,4), (5,3), (6,2) \Rightarrow n = 5$.

$$P(\text{sum} = 8) = \frac{5}{36}.$$

2. Product = 12: favourable pairs $(2,6), (3,4), (4,3), (6,2) \Rightarrow n = 4$.

$$P(\text{product} = 12) = \frac{4}{36} = \frac{1}{9}.$$

3. Let A = "sum is 8" and B = "product is 12". Then $A \cap B = \{(2,6), (6,2)\} \Rightarrow n(A \cap B) = 2$. By the addition law (studied in detail in the next section):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{5}{36} + \frac{4}{36} - \frac{2}{36} = \frac{7}{36}.$$

WORKED EXAMPLE 3 **Balls drawn without replacement**

A box contains 4 red and 6 blue balls. Two balls are drawn without replacement. Find the probability that:

1. both are red;
2. one is red and one is blue (in any order);

3. at least one is red.

SOLUTION

Solution. Total balls = 10. The draws are without replacement, so the second draw depends on the first; we use conditional reasoning (or, equivalently, combinations).

1. $P(\text{both red}) = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$.

Check with combinations: $\frac{\binom{4}{2}}{\binom{10}{2}} = \frac{6}{45} = \frac{2}{15}$. ✓

2. $P(\text{one red, one blue}) = P(RB) + P(BR) = \frac{4}{10} \times \frac{6}{9} + \frac{6}{10} \times \frac{4}{9} = \frac{24}{90} + \frac{24}{90} = \frac{48}{90} = \frac{8}{15}$.

Check with combinations: $\frac{\binom{4}{1}\binom{6}{1}}{\binom{10}{2}} = \frac{24}{45} = \frac{8}{15}$. ✓

3. Using the complement:

$$P(\text{at least one red}) = 1 - P(\text{both blue}) = 1 - \frac{6}{10} \times \frac{5}{9} = 1 - \frac{30}{90} = \frac{60}{90} = \frac{2}{3}.$$

(The complement is faster than adding the three cases RR, RB, BR.)

2.6 Consolidation Exercises

EXERCISE 2 **Classical Probability and Event Types**

1. A fair coin is tossed three times. List the sample space. Find the probability of: (a) exactly two heads; (b) at least two heads; (c) at most one tail.
2. A card is drawn from a standard 52-card pack. Find the probability that it is: (a) a king or a queen; (b) a spade or a face card; (c) neither a heart nor a king.
3. A bag contains 5 white, 4 black and 3 red balls. Three balls are drawn with replacement. Find the probability that: (a) all three are white; (b) exactly two are black; (c) at least one is red.
4. The odds against a student passing a test are 3 : 2. What is the probability that the student passes? If two independent students take the test, what is the probability that both pass?
5. Events A and B are such that $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cup B) = 0.7$. Determine whether A and B are: (a) mutually exclusive; (b) independent.

SOLUTION **Exercise 1**

1. Sample space $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$, $n(S) = 8$.
   - (a) Exactly two heads: $\{HHT, HTH, THH\} \Rightarrow P = \frac{3}{8}$.
   - (b) At least two heads: $\{HHH, HHT, HTH, THH\} \Rightarrow P = \frac{4}{8} = \frac{1}{2}$.
   - (c) At most one tail means at least two heads $\Rightarrow P = \frac{1}{2}$.
2.
   - (a) $P(K \cup Q) = \frac{4+4}{52} = \frac{8}{52} = \frac{2}{13}$ (mutually exclusive).
   - (b) $P(\text{spade} \cup \text{face}) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52} = \frac{11}{26}$ (3 face cards are spades).
   - (c) $P(\text{heart} \cup \text{king}) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$. Complement: $1 - \frac{4}{13} = \frac{9}{13}$.
3. With replacement $\Rightarrow$ independent draws; the bag always contains 12 balls.
   - (a) $P(WWW) = \left(\frac{5}{12}\right)^3 = \frac{125}{1728}$.
   - (b) Exactly two black: $\binom{3}{2}\left(\frac{4}{12}\right)^2\left(\frac{8}{12}\right) = 3 \times \frac{1}{9} \times \frac{2}{3} = \frac{2}{9}$.
   - (c) $P(\text{at least one red}) = 1 - P(\text{no red}) = 1 - \left(\frac{9}{12}\right)^3 = 1 - \frac{27}{64} = \frac{37}{64}$.
4. Odds against $= 3 : 2 \Rightarrow P(\text{pass}) = \frac{2}{3+2} = \frac{2}{5} = 0.4$. Two independent students: $P(\text{both pass}) = (0.4)^2 = 0.16$.
5.
   - (a) If A and B were mutually exclusive, we would have $P(A \cup B) = P(A) + P(B) = 0.9 \neq 0.7$ $\Rightarrow$ they are not mutually exclusive.
   - (b) First compute $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.4 + 0.5 - 0.7 = 0.2$. Since $P(A)P(B) = 0.4 \times 0.5 = 0.2 = P(A \cap B)$, the events are independent.

KEY POINTS **Key Points of This Section**

- Classical probability $P(A) = \frac{n(A)}{n(S)}$ applies only when all outcomes are equally likely.
- $0 \leq P(A) \leq 1$; $P(S) = 1$; $P(\varnothing) = 0$.
- Mutually exclusive events cannot occur together ($A \cap B = \varnothing$).
- Exhaustive events cover the whole sample space; if they are also pairwise disjoint, their probabilities sum to 1.
- Complementary events are mutually exclusive and exhaustive: $P(A') = 1 - P(A)$.
- Use the complement for "at least one" problems: $P(\text{at least one}) = 1 - P(\text{none})$.

- Odds in favour = $p : (1-p)$; odds against = $(1-p) : p$; odds $a : b$ in favour means $P(A) = \frac{a}{a+b}$.
- Mutually exclusive ≠ independent — this distinction is crucial for the coming sections.

DID YOU KNOW? **Did You Know?**

The classical definition of probability, stated by Pierre-Simon Laplace in 1812, was the first rigorous attempt to quantify uncertainty. It works beautifully for games of chance — dice, cards, lotteries — because symmetry guarantees equally likely outcomes. However, it cannot handle questions like "What is the probability that it will rain tomorrow in Yaoundé?" That requires the frequentist (empirical) approach, which we meet in the next section, or the Bayesian approach, which we shall meet at the end of this chapter through Bayes' theorem.

3 EMPIRICAL PROBABILITY AND THE ADDITION LAWS

3.1 From Classical to Empirical Probability

In the previous section we studied **classical (theoretical) probability**: when all outcomes of an experiment are equally likely, $P(A) = \dfrac{n(A)}{n(S)}$. But life is not always a fair die. What is the probability that a drawing pin lands point up? That a newborn baby in Bamenda is a girl? That a candidate passes the GCE at first attempt? No symmetry argument can answer these questions. We must experiment and observe. This leads us to **empirical probability**, and then to two fundamental laws — the **addition laws** — that let us combine probabilities of events.

3.2 Empirical (Experimental) Probability

EXPERIMENT **Tossing a coin 100 times**

Take a coin and toss it 100 times, recording the number of heads after 10, 20, 30, …, 100 tosses. A class in Buea obtained: 7 heads in 10 tosses, then 11, 16, 23, 26, 34, 38, 44, 47, and finally 51 heads in 100 tosses. The relative frequencies of heads are:

$$\frac{7}{10} = 0.70, \quad \frac{11}{20} = 0.55, \quad \frac{16}{30} \approx 0.53, \quad \ldots, \quad \frac{51}{100} = 0.51.$$

The values fluctuate wildly at first, then settle down near 0.5.

DEFINITION **Empirical probability (relative frequency)**

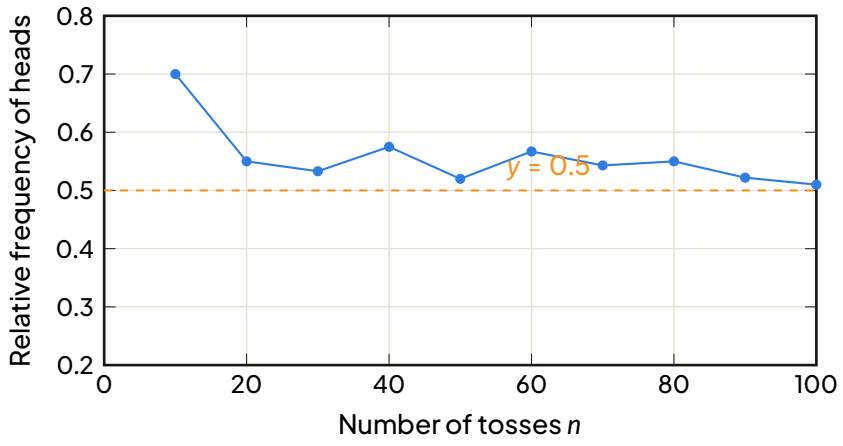
If an experiment is repeated n times under identical conditions and an event A occurs f times, the **relative frequency** of A is

$$f_n(A) = \frac{f}{n} = \frac{\text{number of times } A \text{ occurs}}{\text{number of trials}}.$$

The **empirical (experimental) probability** of A is the value towards which $f_n(A)$ stabilises as n becomes large; in practice we estimate $P(A) \approx f_n(A)$ for large n.

PROPERTY **Stability of relative frequency — informal law of large numbers**

As the number of trials n increases, the relative frequency $f_n(A)$ tends to stabilise around a fixed number, which is the true probability $P(A)$. For a fair coin, $f_n(\text{heads}) \to 0.5$. This is why empirical and theoretical probability agree when the theoretical model is correct. (Proof not examinable.)



Relative frequency of heads against the number of tosses: the fluctuations die out and the frequency converges to the theoretical value 0.5 (dashed line).

EXEMPLE **The drawing pin experiment**

A drawing pin is thrown 200 times and lands point up 118 times. There is no symmetry here, so classical probability is useless. The empirical estimate is

$$P(\text{point up}) \approx \frac{118}{200} = 0.59.$$

Similarly, a health centre in Yaoundé records 520 girls among 1000 births in a year: the empirical probability that a newborn is a girl is $\frac{520}{1000} = 0.52$.

WARNING **Small samples mislead**

With only 10 tosses, a relative frequency of 0.7 tells us almost nothing. Empirical probability is reliable only for a **large** number of trials. Never quote an empirical probability from a handful of trials.

3.3 Addition Law for Mutually Exclusive Events

Recall from Section 2: two events A and B are **mutually exclusive** when they cannot occur together, i.e. $A \cap B = \varnothing$.

PROPERTY **Addition law for mutually exclusive events**

If A and B are mutually exclusive events of a sample space S with equally likely outcomes, then

$$P(A \cup B) = P(A) + P(B).$$

Proof (examinable). Since $A \cap B = \varnothing$, the outcomes of A and B are distinct, so $n(A \cup B) = n(A) + n(B)$. Dividing by $n(S)$:

$$P(A \cup B) = \frac{n(A \cup B)}{n(S)} = \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} = P(A) + P(B).$$

Extension. If $A_1, A_2, \ldots, A_k$ are pairwise mutually exclusive, then

$$P(A_1 \cup A_2 \cup \cdots \cup A_k) = P(A_1) + P(A_2) + \cdots + P(A_k).$$

EXEMPLE **One card from a deck**

A card is drawn from a standard pack of 52. Let A = "draw a king" and B = "draw a queen". These are mutually exclusive, so

$$P(A \cup B) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}.$$

3.4 General Addition Law for Non-Mutually Exclusive Events

What if the events overlap? Suppose 30 students of an Upper Sixth class study French (F) and 25 study English (E), but 12 study both. Then $P(F) + P(E)$ counts the 12 bilingual students twice.

We must subtract the intersection once.

PROPERTY **General addition law**

For any two events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof (examinable). In the count $n(A) + n(B)$, every element of $A \cap B$ is counted twice, so

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

Dividing by $n(S)$ gives the result. When $A \cap B = \varnothing$, $P(A \cap B) = 0$ and we recover the mutually exclusive law.

METHOD **Filling a Venn diagram or two-way table from the inside out**

1. Identify the **innermost region** first: place $n(A \cap B)$ in the overlap.
2. Complete each circle: $n(A \text{ only}) = n(A) - n(A \cap B)$, and similarly for B.
3. Place $n(\text{neither}) = n(S) - n(A \cup B)$ outside the circles.
4. Check that all regions sum to $n(S)$, then read off any probability.

WORKED EXAMPLE 4 **French and English students**

In a class of 60 students, 30 study French, 25 study English and 12 study both. A student is chosen at random. Find the probability that the student studies (a) French or English; (b) neither.

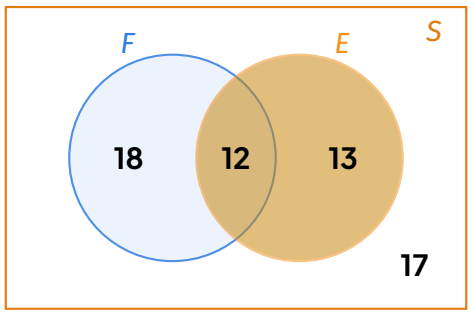
SOLUTION

Solution. Fill the Venn diagram from the inside out: $n(F \cap E) = 12$; French only: $30 - 12 = 18$; English only: $25 - 12 = 13$; neither: $60 - (18 + 12 + 13) = 17$.

$$\text{(a)}\quad P(F \cup E) = P(F) + P(E) - P(F \cap E) = \frac{30}{60} + \frac{25}{60} - \frac{12}{60} = \frac{43}{60}.$$

$$\text{(b)}\quad P(\text{neither}) = \frac{17}{60}.$$

Check with the diagram: $\dfrac{18 + 12 + 13}{60} = \dfrac{43}{60}$. ✓



Venn diagram of the 60 students: $18 + 12 + 13 + 17 = 60$, and $P(F \cup E) = \frac{18+12+13}{60} = \frac{43}{60}$, confirming $P(F) + P(E) - P(F \cap E)$.

The same information can be organised in a **two-way table**:

	English	Not English	Total
French	12	18	30
Not French	13	17	30
Total	25	35	60

WARNING **The double-counting trap**

The most common error at the GCE is writing $P(A \cup B) = P(A) + P(B)$ when A and B overlap. This counts $A \cap B$ twice and can even give a probability greater than 1 — an immediate sign of error. Always ask first: can the two events happen together? If yes, subtract $P(A \cap B)$.

EXERCISE 3 **Survey at a market**

Of 80 shoppers interviewed in Douala, 45 bought plantains, 38 bought tomatoes and 20 bought both. A shopper is chosen at random. Find the probability that the shopper bought (a) plantains or tomatoes; (b) tomatoes only; (c) neither.

SOLUTION **Exercise 1**

Both: 20; plantains only: $45 - 20 = 25$; tomatoes only: $38 - 20 = 18$; neither: $80 - (25 + 20 + 18) = 17$. (a) $P = \frac{45}{80} + \frac{38}{80} - \frac{20}{80} = \frac{63}{80}$. (b) $P = \frac{18}{80} = \frac{9}{40}$. (c) $P = \frac{17}{80}$.

KEY POINTS **Key points**

- Empirical probability: $P(A) \approx \dfrac{\text{number of times } A \text{ occurs}}{\text{number of trials}}$, reliable only for many trials; relative frequency stabilises towards the true probability (law of large numbers).
- Mutually exclusive: $P(A \cup B) = P(A)+P(B)$; this extends to any number of pairwise mutually exclusive events.
- General case: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Organise overlapping data with a Venn diagram or two-way table, filled from the intersection outwards.

4 MULTIPLICATION LAW, CONDITIONAL PROBABILITY AND INDEPENDENCE

In the previous section we studied empirical probability and the addition laws, which answer the question: "What is the probability that A **or** B occurs?" In real life, however, we very often need to answer a different question: "What is the probability that A **and** B occur?" or even "What is the probability of A knowing that B has already happened?" Think of a doctor in Bamenda who knows a patient has fever (B) and wants the probability of malaria (A); or a student who has already drawn one red pen from a box and asks for the probability of drawing a second red pen. These questions lead us to **conditional probability**, the **multiplication law** and the fundamental notion of **independence**.

4.1 Conditional Probability: Probability with Extra Information

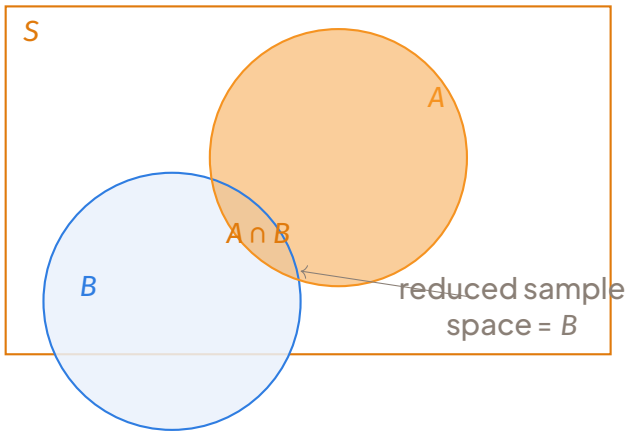
Intuition first.

A school in Buea has 200 Upper Sixth students, classified by sex and by the field they intend to study:

	Science (A)	Arts (A')	Total
Boys (B)	60	30	90
Girls (B')	50	60	110
Total	110	90	200

A student is chosen at random. The probability that the student intends to study science is $P(A) = \dfrac{110}{200} = 0.55$. But suppose we are told that the chosen student is a boy. This extra information changes everything: the whole school is no longer our universe. We now pick from the 90 boys only, of whom 60 choose science, so the probability becomes $\dfrac{60}{90} = \dfrac{2}{3} \approx 0.667$. We write this as $P(A \mid B)$, read "the probability of A **given** B".

The key idea is that knowing B has occurred **reduces the sample space** from S to B. Inside this reduced sample space, the part where A also occurs is exactly $A \cap B$.



Reduced sample space: once B is known to have occurred, only the shaded disc B remains possible, and $P(A \mid B)$ is the proportion of B covered by $A \cap B$.

For equally likely outcomes, counting inside the reduced sample space gives

$$P(A \mid B) = \frac{n(A \cap B)}{n(B)} = \frac{n(A \cap B)/n(S)}{n(B)/n(S)} = \frac{P(A \cap B)}{P(B)}.$$

This motivates the general definition.

DEFINITION **Conditional probability**

Let A and B be two events of a sample space S with $P(B) > 0$. The **conditional probability** of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

It is the probability that A occurs knowing that B has occurred. Geometrically, it is the proportion of the event B that is covered by $A \cap B$.

EXEMPLE **Computing from the contingency table**

Using the table of 200 students above, with A = “intends to study science” and B = “is a boy”:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{60/200}{90/200} = \frac{60}{90} = \frac{2}{3}.$$

Similarly, the probability that a student is a boy given that he or she intends to study sci-

ence is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{60/200}{110/200} = \frac{60}{110} = \frac{6}{11}.$$

Note that $P(A \mid B) = \frac{2}{3} \neq \frac{6}{11} = P(B \mid A)$.

WARNING **$P(A \mid B)$ and $P(B \mid A)$ are generally different**

The word "given" tells you which event becomes the new sample space (the denominator). $P(A \mid B)$ uses $P(B)$ in the denominator; $P(B \mid A)$ uses $P(A)$. Confusing the two is one of the most frequent errors at the GCE. Always read the sentence slowly and underline what is known.

WORKED EXAMPLE 5 **Conditional probability from a Venn diagram**

In a class, $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$. Find (a) $P(A \mid B)$; (b) $P(B \mid A)$; (c) $P(A \mid B')$.

SOLUTION

Solution.

1. $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.2}{0.4} = 0.5.$

2. $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{0.2}{0.5} = 0.4.$

3. First $P(A \cap B') = P(A) - P(A \cap B) = 0.5 - 0.2 = 0.3$, and $P(B') = 1 - 0.4 = 0.6$. Hence

$$P(A \mid B') = \frac{P(A \cap B')}{P(B')} = \frac{0.3}{0.6} = 0.5.$$

4.2 The Multiplication Law

Rearranging the definition of conditional probability (multiplying both sides by $P(B)$) immediately gives the rule for the probability of an intersection.

PROPERTY **Multiplication law of probability**

For any two events A and B with $P(A) > 0$ and $P(B) > 0$,

$$P(A \cap B) = P(A) \times P(B \mid A) = P(B) \times P(A \mid B).$$

(Proof examinable: it is a direct rearrangement of the definition $P(A \mid B) = P(A \cap B)/P(B)$.)

The law extends naturally to three or more events, for example

$$P(A \cap B \cap C) = P(A)\,P(B \mid A)\,P(C \mid A \cap B).$$

EXEMPLE **Drawing without replacement**

A box contains 5 red and 3 blue biros. Two biros are drawn at random without replacement. What is the probability that both are red?
Let R_1 = "first biro red" and R_2 = "second biro red". Then

$$P(R_1 \cap R_2) = P(R_1) \times P(R_2 \mid R_1) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}.$$

After the first red biro is removed, only 4 red biros remain out of 7: this is exactly the conditional probability $P(R_2 \mid R_1)$.

4.3 Independent Events

Intuition.

Sometimes knowing that B has occurred gives no new information about A: the probability of A stays the same. Tossing a coin and rolling a die: knowing the coin shows Head does not change the probability of obtaining a 6. Such events are called **independent**.

If A and B are independent, then $P(A \mid B) = P(A)$, and the multiplication law becomes $P(A \cap B) = P(A)\,P(B)$. We take this simpler identity as the definition, because it is symmetric and works even when one probability is zero.

DEFINITION **Independent events**

Two events A and B are **independent** if and only if

$$P(A \cap B) = P(A) \times P(B).$$

When $P(A) > 0$ and $P(B) > 0$, this is equivalent to each of the conditions

$$P(A \mid B) = P(A) \qquad \text{and} \qquad P(B \mid A) = P(B).$$

Otherwise the events are **dependent**.

PROPERTY **Independence of complements**

If A and B are independent, then so are the pairs (A, B'), (A', B) and (A', B').

Proof (instructive). Since $A = (A \cap B) \cup (A \cap B')$, a union of disjoint events,

$$P(A \cap B') = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)\big(1 - P(B)\big) = P(A)P(B').$$

Hence A and B' are independent. Applying the same argument to A' and B, and then to A' and B', completes the proof. ■

METHOD **Testing independence numerically**

To decide whether two events A and B are independent from data or given probabilities:

1. Compute (or read off) $P(A)$, $P(B)$ and $P(A \cap B)$.
2. Compute the product $P(A) \times P(B)$.
3. Compare: if $P(A \cap B) = P(A)P(B)$, the events are independent; otherwise they are dependent.
4. Equivalently, check whether $P(A \mid B) = P(A)$.

WORKED EXAMPLE 6 **Are the events independent?**

Using the table of 200 students: A = "intends to study science", B = "is a boy". Test whether A and B are independent.

SOLUTION

Solution. From the table:

$$P(A) = \frac{110}{200} = 0.55, \qquad P(B) = \frac{90}{200} = 0.45, \qquad P(A \cap B) = \frac{60}{200} = 0.30.$$

Now $P(A) \times P(B) = 0.55 \times 0.45 = 0.2475 \neq 0.30 = P(A \cap B)$.
Therefore A and B are not independent: the intention to study science depends on sex in this school. Equivalently, $P(A \mid B) = \frac{2}{3} \approx 0.667 \neq 0.55 = P(A)$.

EXEMPLE **An independent situation**

A fair coin is tossed and a fair die is rolled. Let A = "Head" and B = "a six". Then

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{6}, \quad P(A \cap B) = \frac{1}{12} = \frac{1}{2} \times \frac{1}{6}.$$

The events are independent, as expected: the coin cannot influence the die.

4.4 Mutually Exclusive versus Independent: a Classic Trap

These two notions are constantly confused by candidates, yet they are almost opposite ideas:

- **Mutually exclusive** (disjoint): $A \cap B = \varnothing$, i.e. $P(A \cap B) = 0$. The occurrence of one prevents the other.

- **Independent**: $P(A \cap B) = P(A)P(B)$. The occurrence of one gives no information about the other.

WARNING **Disjoint events with positive probabilities are never independent**

If A and B are mutually exclusive with $P(A) > 0$ and $P(B) > 0$, then

$$P(A \cap B) = 0 \neq P(A)P(B) > 0,$$

so A and B are dependent. Indeed, if B occurs then A cannot occur: $P(A \mid B) = 0 \neq P(A)$. Knowing B completely determines A — the strongest possible dependence! Counter-example to keep in mind. Drawing one card from a pack: A = "king", B = "queen". They are mutually exclusive but clearly not independent. Conversely, A = "king" and C = "heart" are independent ($P(A \cap C) = \frac{1}{52} = \frac{4}{52} \times \frac{13}{52}$) but not mutually exclusive (the king of hearts exists). Never write $P(A \cap B) = P(A)P(B)$ for disjoint events, and never write $P(A \cup B) = P(A) + P(B)$ for events that overlap.

4.5 Exhaustive Events and a Preview of Total Probability

Recall that events $B_1, B_2, \ldots, B_n$ form a **partition** of S (they are exhaustive and mutually exclusive) when every outcome belongs to exactly one of them. Then any event A can be split into disjoint pieces:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_n).$$

Applying the addition law for mutually exclusive events, then the multiplication law to each piece:

$$P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(B_i)\,P(A \mid B_i).$$

PROPERTY **Law of total probability — preview**

If $B_1, B_2, \ldots, B_n$ form a partition of S with $P(B_i) > 0$ for all i, then for any event A,

$$P(A) = P(B_1)P(A \mid B_1) + P(B_2)P(A \mid B_2) + \cdots + P(B_n)P(A \mid B_n).$$

This result will be proved and used fully in Section 7, where it becomes the engine of Bayes' theorem.

EXEMPLE **Splitting over a partition**

A market seller in Douala buys tomatoes from two farms: 60% from farm B_1 and 40% from farm B_2. 5% of farm B_1's tomatoes are spoilt and 10% of farm B_2's are spoilt. Let A = "a randomly picked tomato is spoilt". Then

$$P(A) = P(B_1)P(A \mid B_1) + P(B_2)P(A \mid B_2) = (0.6)(0.05) + (0.4)(0.10) = 0.03 + 0.04 = 0.07.$$

So 7% of the tomatoes are spoilt overall — computed by splitting A over the partition $\{B_1, B_2\}$.

KEY POINTS **Key points of this section**

- Conditional probability: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, $P(B) > 0$ — the sample space is reduced to B.
- Multiplication law: $P(A \cap B) = P(A)\,P(B \mid A) = P(B)\,P(A \mid B)$.
- Independence: $P(A \cap B) = P(A)P(B)$, equivalently $P(A \mid B) = P(A)$; complements of independent events remain independent.
- Mutually exclusive ≠ independent: disjoint events with positive probabilities are always dependent.
- Total probability: if $B_1, \dots, B_n$ partition S, then $P(A) = \sum_i P(B_i)P(A \mid B_i)$.

EXERCISE 4 **Practice**

A bag contains 4 white and 6 black balls. Two balls are drawn at random without replacement. Let A = "first ball white" and B = "second ball white".

1. Compute $P(A)$, $P(B \mid A)$ and $P(A \cap B)$.
2. Using the law of total probability over the partition $\{A, A'\}$, compute $P(B)$.
3. Are A and B independent? Justify.

SOLUTION **Exercise 1**

1. $P(A) = \dfrac{4}{10} = \dfrac{2}{5}$; $P(B \mid A) = \dfrac{3}{9} = \dfrac{1}{3}$; $P(A \cap B) = \dfrac{2}{5} \times \dfrac{1}{3} = \dfrac{2}{15}$.
2. $P(B) = P(A)P(B \mid A) + P(A')P(B \mid A') = \dfrac{4}{10} \times \dfrac{3}{9} + \dfrac{6}{10} \times \dfrac{4}{9} = \dfrac{12 + 24}{90} = \dfrac{2}{5}$.
3. $P(A)P(B) = \dfrac{2}{5} \times \dfrac{2}{5} = \dfrac{4}{25} \neq \dfrac{2}{15} = P(A \cap B)$, so A and B are dependent (drawing without replacement creates dependence).

5 TREE DIAGRAMS AND SELECTION WITH OR WITHOUT REPLACEMENT

In the previous section we met the multiplication law $P(A \cap B) = P(A) \times P(B \mid A)$ and the notion of independence. These ideas are powerful, but when an experiment has several stages — two coins tossed one after another, two balls drawn from a bag, a machine tested on three consecutive days — it becomes difficult to keep track of all the cases in one's head. The **tree diagram** is the picture that organises the multiplication law for us: it displays every possible route through the experiment, with the probability of each step written on the branch, so that no case is forgotten and no probability is misapplied.

5.1 Constructing a Tree Diagram

DEFINITION **Tree diagram**

A **tree diagram** for a multi-stage experiment is a diagram in which:

- each stage of the experiment is represented by a level of branching;
- from each node, one branch is drawn for each possible outcome of that stage;
- the probability of that outcome (given the history so far) is written on the branch;
- the final outcome of each complete route is written at the end of the path, together with its probability.

PROPERTY **Rules of the tree**

1. Node rule: the probabilities on all branches leaving the same node sum to 1.
2. Path rule (multiplication): the probability of a complete path is the product of the probabilities on its branches.
3. Event rule (addition): the probability of an event is the sum of the probabilities of all final outcomes that satisfy the event.
4. Check: the probabilities of all final outcomes of the tree sum to 1.

The path rule is exactly the multiplication law $P(A \cap B) = P(A)\,P(B \mid A)$; the event rule is the addition law for mutually exclusive final outcomes. (Justification examinable as an application of those two laws.)

METHOD **Solving a probability problem with a tree**

1. Identify the stages of the experiment and the possible outcomes at each stage.
2. Draw the tree; write on each branch the probability of that outcome given what has already happened (conditional probabilities on second and later levels).
3. Multiply along each path to obtain the probability of each final outcome.
4. Check: the final probabilities sum to 1.
5. Add the probabilities of exactly those final outcomes that satisfy the required event.

WARNING **Multiply along a path — never add**

A very frequent error is to add the probabilities written along a single path. Along a path, events happen one after another ("and"), so we multiply. We only add when combining different final outcomes ("or").

5.2 Selection With and Without Replacement

Many GCE problems concern objects drawn from a collection: balls from a bag, cards from a pack, items from a production line. The crucial question is: **is the first object put back before the second draw?**

DEFINITION **With and without replacement**

- With replacement: the object drawn is returned before the next draw. The composition of the collection is unchanged, so the probabilities are the same at every stage and the stages are **independent**.
- Without replacement: the object drawn is kept out. The composition changes, so the probabilities at the next stage must be updated; the stages are **dependent**, and second-level branch probabilities are conditional probabilities $P(\text{second} \mid \text{first})$.

METHOD **Updating probabilities without replacement**

After a draw without replacement, at the next stage:

1. decrease by 1 the number of objects in the category just drawn;
2. decrease by 1 the total number of objects;

3. leave all other categories unchanged.

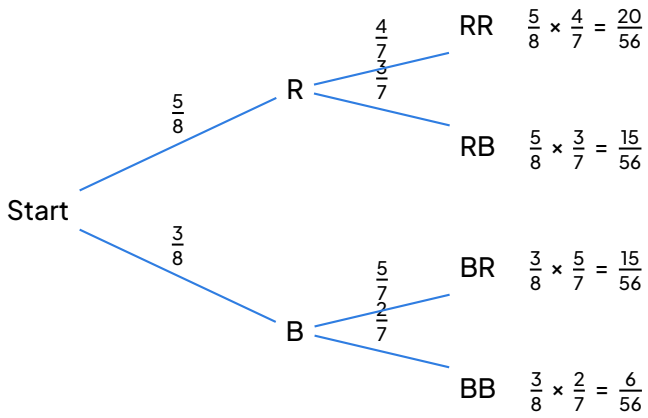
WARNING **The most common GCE error**

With replacement the denominators stay the same; without replacement they change. Drawing from a bag of 8 balls: with replacement, every denominator is 8; without replacement, the second-stage denominators are all 7. Mixing these up destroys the whole calculation — always ask first: is it replaced?

5.3 Worked Example: Two Balls Without Replacement

EXEMPLE **Bag of 5 red and 3 blue balls**

A bag contains 5 red and 3 blue balls. Two balls are drawn at random, one after the other, without replacement. First draw: $P(R) = \frac{5}{8}$, $P(B) = \frac{3}{8}$. If red was drawn first, 4 red and 3 blue remain, so $P(R \mid R) = \frac{4}{7}$, $P(B \mid R) = \frac{3}{7}$. If blue was drawn first, 5 red and 2 blue remain, so $P(R \mid B) = \frac{5}{7}$, $P(B \mid B) = \frac{2}{7}$.



Two draws without replacement from 5 red and 3 blue balls. Check: $\frac{20+15+15+6}{56} = 1$.

From the tree: $P(\text{both red}) = \frac{20}{56} = \frac{5}{14}$; $P(\text{one of each colour}) = \frac{15}{56} + \frac{15}{56} = \frac{15}{28}$; $P(\text{at least one blue}) = 1 - P(RR) = 1 - \frac{20}{56} = \frac{9}{14}$ (complement shortcut).

The tree also answers conditional questions directly. For example, “given that the second ball is blue, what is the probability that the first was red?” The second ball is blue on paths RB and

BB, with total probability $\frac{15}{56} + \frac{6}{56} = \frac{21}{56}$; only RB has first red, so

$$P(\text{first red} \mid \text{second blue}) = \frac{P(RB)}{P(RB) + P(BB)} = \frac{15/56}{21/56} = \frac{15}{21} = \frac{5}{7}.$$

5.4 Three-Stage Experiments: Three Coin Tosses

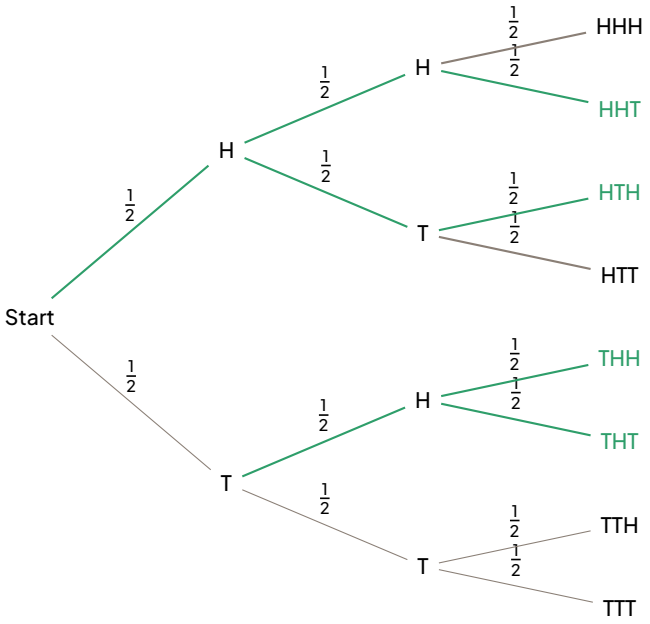
WORKED EXAMPLE 7 **Exactly two heads in three tosses**

A fair coin is tossed three times. Find the probability of obtaining exactly two heads.

SOLUTION

Each toss gives H or T with probability $\frac{1}{2}$; the tosses are independent, so every complete path has probability $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$. The paths with exactly two heads are HHT, HTH, THH — three paths. Hence

$$P(\text{exactly two heads}) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}.$$



Three coin tosses. The green paths (HHT, HTH, THH) give exactly two heads: $P = \frac{3}{8}$.

5.5 "At least one" and "exactly one" from Trees

METHOD **The complement shortcut for "at least one"**

To find P(at least one success):

1. locate the single path with no success (e.g. all failures);
2. compute its probability by multiplying along that path;
3. subtract from 1: P(at least one) = 1 − P(none).

This avoids adding many paths and is much faster in the examination.

EXEMPLE **Quality control**

A machine produces items of which 10% are defective. Three items are tested independently (equivalent to sampling with replacement). Then

$$P(\text{at least one defective}) = 1 - P(\text{none defective}) = 1 - (0.9)^3 = 1 - 0.729 = 0.271.$$

For "exactly one defective", add the three paths DGG, GDG, GGD: $3 \times (0.1)(0.9)^2 = 0.243$.

WARNING **Check your tree before using it**

Before reading off any answer, verify the two quick checks: (i) branches from each node sum to 1; (ii) all final outcomes sum to 1. If either fails, a branch probability is wrong — usually a denominator that was not updated after a draw without replacement.

DID YOU KNOW? **Did you know?**

Tree diagrams are the everyday tool of decision analysis: doctors use them to combine the probabilities of successive test results, and engineers use them (as "fault trees") to estimate the reliability of systems such as the power grid. The same multiply-along, add-across logic you use for balls in a bag drives software that assesses risks in aviation and medicine.

EXERCISE 5

A box contains 4 white and 6 black balls. Two balls are drawn at random.

1. Draw the tree and find P(both black) when the draw is with replacement.
2. Repeat when the draw is without replacement.

3. In each case, find P(the two balls are of different colours).

SOLUTION **Exercise 1**

With replacement: denominators stay 10. $P(BB) = \frac{6}{10} \times \frac{6}{10} = \frac{9}{25}$. $P(\text{different}) = \frac{6}{10} \cdot \frac{4}{10} + \frac{4}{10} \cdot \frac{6}{10} = \frac{12}{25}$.
Without replacement: denominators become 9. $P(BB) = \frac{6}{10} \times \frac{5}{9} = \frac{1}{3}$. $P(\text{different}) = \frac{6}{10} \cdot \frac{4}{9} + \frac{4}{10} \cdot \frac{6}{9} = \frac{48}{90} = \frac{8}{15}$.

EXERCISE 6

In a certain school, 60% of the candidates enter for Mathematics and 40% for Biology only at a given session; independently, each candidate passes with probability 0.7. A candidate is chosen at random.

1. Draw a two-stage tree (subject, then result) and find the probability that the candidate entered for Mathematics and passed.
2. Find the probability that the candidate passed.
3. Given that the candidate passed, find the probability that she entered for Mathematics.

SOLUTION **Exercise 2**

First level: $P(M) = 0.6$, $P(B) = 0.4$. Second level: $P(\text{pass} \mid M) = P(\text{pass} \mid B) = 0.7$, $P(\text{fail}) = 0.3$ on each branch.

1. $P(M \cap \text{pass}) = 0.6 \times 0.7 = 0.42$.
2. $P(\text{pass}) = 0.6 \times 0.7 + 0.4 \times 0.7 = 0.42 + 0.28 = 0.70$ (as expected, since the pass rate is the same for both subjects).
3. $P(M \mid \text{pass}) = \dfrac{0.42}{0.70} = 0.6$.

KEY POINTS **Key points**

- One branch per outcome; probability on the branch; outcome at the end.

- Multiply along a path; add across final outcomes; all final probabilities sum to 1.
- With replacement: probabilities constant, stages independent. Without replacement: decrease the drawn category and the total by 1; second-level branches are conditional probabilities.
- "At least one": use 1 − P(none). "Exactly one": add the relevant paths.
- Conditional questions from a tree: $P(A \mid B) = \dfrac{P(\text{paths in } A \cap B)}{P(\text{paths in } B)}$.

6 COUNTING TECHNIQUES IN PROBABILITY

In the previous section we used tree diagrams to analyse selections made with or without replacement. Trees are wonderful for two or three stages, but they collapse under their own weight very quickly: drawing a tree for choose 3 balls from 10 already requires $10 \times 9 \times 8 = 720$ terminal branches. Yet the GCE examiner will happily ask you for the probability that a committee of 5 chosen from 12 people contains exactly 2 women, or that a sample of 4 items from a crate of 50 contains at least one defective. For such problems we need a faster way to count outcomes without listing them. That is exactly what **counting techniques** — the multiplication principle, permutations and combinations — provide. This section shows how they turn impossible listing problems into one-line calculations.

6.1 Why Counting Matters

Recall the classical definition of probability from earlier in this chapter: when all outcomes of a sample space S are equally likely,

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}.$$

Everything therefore reduces to counting two sets: the favourable outcomes and all outcomes. When $n(S) = 6$ (one die) or $n(S) = 36$ (two dice), listing is feasible. But consider:

- choosing 3 balls from a bag of 10: the number of possible selections is 120;
- forming a committee of 5 from 12 candidates: 792 possible committees;
- dealing a hand of 5 cards from a pack of 52: 2 598 960 possible hands.

No candidate can list these in an examination. The whole art of this section is to compute $n(A)$ and $n(S)$ by formula, and — crucially — to count both with the same rule.

WARNING **Count consistently**

Whatever counting rule you use for the numerator $n(A)$, you must use the same rule for

the denominator $n(S)$. Mixing ordered counting above with unordered counting below (or vice versa) is the single most common source of wrong answers in this topic.

6.2 The Multiplication Principle

EXPERIMENT **Counting routes from Bamenda to Buea**

Suppose a student travels from Bamenda to Buea, passing through Douala. There are 3 possible coaches from Bamenda to Douala, and 4 possible coaches from Douala to Buea. How many complete journeys are possible? For each of the 3 first-leg choices there are 4 second-leg choices, giving $3 \times 4 = 12$ journeys. No list is needed: the choices multiply.

PROPERTY **Multiplication principle of counting**

If an experiment is performed in stages, where the first stage can occur in n_1 ways, the second in n_2 ways, …, and the k-th in n_k ways, then the total number of possible outcomes is

$$n_1 \times n_2 \times \cdots \times n_k.$$

This holds whether the choices at each stage are independent or not, provided the number of options at each stage does not depend on the specific choices made earlier (only the set of available options may change, but its size is fixed).

EXEMPLE **Number plates**

A number plate consists of 2 letters followed by 3 digits. Assuming repetition is allowed, the number of plates is

$$26 \times 26 \times 10 \times 10 \times 10 = 676\,000.$$

If repetition is not allowed among the digits, the count becomes $26 \times 26 \times 10 \times 9 \times 8 = 486\,720$.

6.3 Permutations: When Order Matters

Suppose 8 athletes contest the 100 m final at the school sports day in Yaoundé. In how many ways can the gold, silver and bronze medals be awarded? There are 8 choices for gold, then 7 for silver, then 6 for bronze: $8 \times 7 \times 6 = 336$ ways. Here order matters: "Ngong gold, Etame silver" is a different outcome from "Etame gold, Ngong silver". Such an ordered arrangement is called a **permutation**.

DEFINITION **Permutation**

A permutation of r objects chosen from n distinct objects is an ordered selection. The number of such permutations is

$$ {}^{n}P_{r} = \frac{n!}{(n-r)!} = n(n-1)(n-2)\cdots(n-r+1). $$

In particular, the number of arrangements of all n objects is ${}^{n}P_{n} = n!$.

The formula is a direct consequence of the multiplication principle: n choices for the first position, $n-1$ for the second, down to $n-r+1$ for the r-th position.

EXEMPLE **Arranging digits**

How many 3–digit numbers can be formed from the digits 1, 2, 3, 4, 5 without repetition? This is an ordered selection of 3 digits from 5:

$$ {}^{5}P_{3} = \frac{5!}{2!} = 5 \times 4 \times 3 = 60. $$

Hence, if one such number is picked at random, the probability that it is a specific number, say 352, is $\frac{1}{60}$.

EXEMPLE **Seating arrangements**

In how many ways can 6 students be seated in a row? ${}^{6}P_{6} = 6! = 720$. If two particular friends, Amina and Bih, must sit together, treat the pair as one block: $5!$ arrangements of the block with the other 4 students, times $2!$ internal arrangements of the pair, giving $5! \times 2! = 240$. Therefore, if the 6 students sit at random,

$$ P(\text{Amina and Bih together}) = \frac{240}{720} = \frac{1}{3}. $$

6.4 Combinations: When Order Does Not Matter

Now change the question: from 8 athletes, in how many ways can we select 3 to represent the school — with no ranking among them? Choosing ”Ngong, Etame, Ali” is the same selection as ”Ali, Ngong, Etame”. Order no longer matters. Each unordered group of 3 corresponds to $3! = 6$ ordered permutations, so we must divide the permutation count by $3!$.

DEFINITION **Combination**

A combination of r objects chosen from n distinct objects is an unordered selection. The number of such combinations is

$$^nC_r = \binom{n}{r} = \frac{n!}{r!\,(n-r)!} = \frac{^nP_r}{r!}.$$

PROPERTY **Symmetry of combinations**

$$^nC_r = {}^nC_{n-r}.$$

Choosing the r objects to take is the same as choosing the $n - r$ objects to leave behind.

WARNING **Use the symmetry to save time**

Never compute $^{50}C_{47}$ directly: write $^{50}C_{47} = {}^{50}C_3 = \frac{50 \times 49 \times 48}{3 \times 2 \times 1} = 19\,600$. In the examination, always replace r by $n - r$ when $r > \frac{n}{2}$. This avoids long factorial arithmetic and reduces the risk of slips.

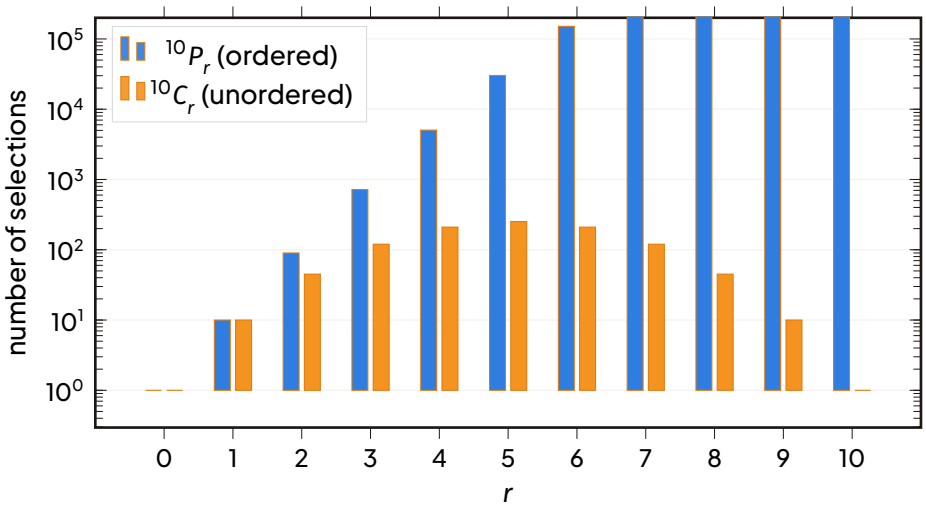
METHOD **Permutation or combination?**

1. Read the question and ask: does the order of selection distinguish two outcomes?
2. If yes (rankings, arrangements, codes, positions, first/second/third prizes): use ^nP_r.
3. If no (committees, groups, hands of cards, samples of items): use ^nC_r.
4. Count the favourable outcomes $n(A)$ and the total outcomes $n(S)$ with the same rule, then form $P(A) = \frac{n(A)}{n(S)}$.

WARNING **Never mix the two rules**

If you count favourable committees with combinations (unordered) but count the total with permutations (ordered), the ratio is meaningless. For example, for "3 balls from 10", $\frac{^6C_3}{^{10}P_3} = \frac{20}{720}$ is wrong; the correct value is $\frac{^6C_3}{^{10}C_3} = \frac{20}{120} = \frac{1}{6}$. Both counts must treat order in

the same way.



Comparison of $^{10}P_r$ and $^{10}C_r$ for $r = 0, 1, \ldots, 10$ (logarithmic scale). Ordered counts are always at least as large as unordered counts, and exceed them by the factor $r!$ as soon as $r \geq 2$.

6.5 Computing Probabilities with Combinations

The standard model is a population split into categories (red/black balls, men/women, defective/good items) from which a sample is drawn without replacement and without order. The total number of samples is a combination; the favourable samples are counted category by category using the multiplication principle, and the counts are then multiplied.

WORKED EXAMPLE 8 **Balls from a bag**

A bag contains 6 red balls and 4 black balls. Three balls are drawn at random, without replacement. Find the probability that all three are red.

SOLUTION

Order does not matter, so we use combinations throughout. Total outcomes: the number of ways of choosing 3 balls from 10 is

$$n(S) = {}^{10}C_3 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120.$$

Favourable outcomes: the number of ways of choosing 3 red balls from the 6 red ones is

$$n(A) = {}^{6}C_3 = \frac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20.$$

Therefore

$$P(\text{all red}) = \frac{{}^{6}C_{3}}{{}^{10}C_{3}} = \frac{20}{120} = \frac{1}{6}.$$

Check with conditional probability (as in the previous section): $\frac{6}{10} \times \frac{5}{9} \times \frac{4}{8} = \frac{120}{720} = \frac{1}{6}$. The two methods agree, as they must.

WORKED EXAMPLE 9 **A committee with exactly 2 women**

A committee of 3 people is to be chosen at random from 4 men and 6 women. Find the probability that the committee contains exactly 2 women.

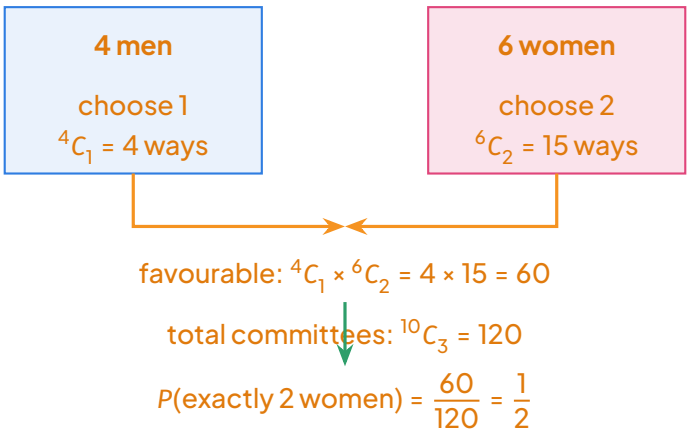
SOLUTION

Total committees: $n(S) = {}^{10}C_{3} = 120$.
Favourable committees: choose 2 women from 6 and 1 man from 4. By the multiplication principle,

$$n(A) = {}^{6}C_{2} \times {}^{4}C_{1} = 15 \times 4 = 60.$$

Hence

$$P(\text{exactly 2 women}) = \frac{{}^{6}C_{2}\,{}^{4}C_{1}}{{}^{10}C_{3}} = \frac{60}{120} = \frac{1}{2}.$$



The committee-selection problem: favourable selections are built category by category ($^{4}C_{1}$ from the men, $^{6}C_{2}$ from the women) and multiplied; the total number of committees is $^{10}C_{3}$.

6.6 "At Least One": Use the Complement

Phrases such as at least one, at least two, no fewer than signal that direct counting is tedious (several cases to add) but the complement is a single easy case.

METHOD **Complement counting for "at least one"**

1. Identify the complementary event: "at least one defective" becomes "no defective" (all good).
2. Count the complement with combinations: $n(\text{none}) = {}^{g}C_{r}$, where g is the number of good items.
3. Compute $P(\text{at least one}) = 1 - \dfrac{n(\text{none})}{n(S)}$.

WORKED EXAMPLE 10 **At least one defective item**

A box contains 20 electronic components, of which 3 are defective. A sample of 4 components is selected at random for testing. Find the probability that the sample contains at least one defective component.

SOLUTION

The total number of samples is

$$n(S) = {}^{20}C_{4} = \frac{20 \times 19 \times 18 \times 17}{4 \times 3 \times 2 \times 1} = 4845.$$

The complement of "at least one defective" is "no defective", i.e. all 4 chosen from the 17 good components:

$$n(\text{no defective}) = {}^{17}C_{4} = \frac{17 \times 16 \times 15 \times 14}{4 \times 3 \times 2 \times 1} = 2380.$$

Therefore

$$P(\text{at least one defective}) = 1 - \frac{{}^{17}C_{4}}{{}^{20}C_{4}} = 1 - \frac{2380}{4845} = \frac{2465}{4845} = \frac{29}{57} \approx 0.509.$$

Note how much shorter this is than adding the three cases "exactly 1, exactly 2, exactly 3 defective".

6.7 Linking Counting Back to Tree Diagrams

Counting formulas are not magic: on small cases they can be verified with the tree diagrams of the previous section, and this cross-check is excellent examination practice.

EXPERIMENT **Verifying ${}^{4}C_{2}$ with a tree**

A bag holds 2 red balls R_1, R_2 and 2 black balls B_1, B_2. Two balls are drawn without replacement. The combination formula predicts ${}^4C_2 = 6$ possible (unordered) pairs: R_1R_2, R_1B_1, R_1B_2, R_2B_1, R_2B_2, B_1B_2. Drawing the full two-stage tree gives $4 \times 3 = 12$ ordered outcomes; grouping each pair of reversed duplicates (R_1R_2 with R_2R_1, etc.) recovers exactly the 6 unordered pairs, since $12 \div 2! = 6$. This is precisely why ${}^nC_r = {}^nP_r / r!$.
Moreover, the probability of drawing one ball of each colour can be computed two ways:

- by combinations: $\dfrac{{}^2C_1\,{}^2C_1}{{}^4C_2} = \dfrac{4}{6} = \dfrac{2}{3}$;
- by the tree: $P(RB) + P(BR) = \dfrac{2}{4}\cdot\dfrac{2}{3} + \dfrac{2}{4}\cdot\dfrac{2}{3} = \dfrac{1}{3} + \dfrac{1}{3} = \dfrac{2}{3}$.

The agreement confirms that combinations are simply trees with the folding done algebraically.

KEY POINTS Key points of this section

- Classical probability is a counting problem: $P(A) = \dfrac{n(A)}{n(S)}$; when listing is impossible, count by formula.
- Multiplication principle: stages multiply, $n_1 \times n_2 \times \cdots \times n_k$.
- Order matters ⇒ permutations: ${}^nP_r = \dfrac{n!}{(n-r)!}$.
- Order does not matter ⇒ combinations: ${}^nC_r = \dfrac{n!}{r!(n-r)!}$.
- Use the same counting rule in the numerator and the denominator.
- For "at least one", count the complement: $P = 1 - \dfrac{{}^gC_r}{{}^nC_r}$.
- Simplify with ${}^nC_r = {}^nC_{n-r}$.

EXERCISE 7 Practice

1. A committee of 4 is chosen at random from 5 teachers and 7 students. Find the probability that the committee contains (a) exactly 2 teachers; (b) at least one teacher.
2. Six books are arranged at random on a shelf. Find the probability that two particular books are at the two ends.
3. A lot of 25 phones contains 4 defective ones. Three phones are tested at random. Find

the probability that (a) none is defective; (b) at least one is defective.

4. From the digits $1, 2, 3, 4, 5, 6$, a 4–digit code is formed without repetition. Find the probability that the code is even.

SOLUTION **Exercise 1**

1. $n(S) = {}^{12}C_4 = 495$. (a) $P(\text{exactly 2 teachers}) = \dfrac{{}^5C_2\,{}^7C_2}{{}^{12}C_4} = \dfrac{10 \times 21}{495} = \dfrac{210}{495} = \dfrac{14}{33}$. (b) $P(\text{at least one teacher}) = 1 - \dfrac{{}^7C_4}{{}^{12}C_4} = 1 - \dfrac{35}{495} = \dfrac{460}{495} = \dfrac{92}{99}$.
2. Total arrangements: $6! = 720$. Favourable: place the two books at the ends ($2!$ ways) and arrange the rest ($4!$ ways): $2! \times 4! = 48$. Hence $P = \dfrac{48}{720} = \dfrac{1}{15}$.
3. (a) $P(\text{none defective}) = \dfrac{{}^{21}C_3}{{}^{25}C_3} = \dfrac{1330}{2300} = \dfrac{133}{230}$. (b) $P(\text{at least one}) = 1 - \dfrac{133}{230} = \dfrac{97}{230}$.
4. Total codes: ${}^6P_4 = 360$. Even codes end in 2, 4 or 6: 3 choices for the last digit, then ${}^5P_3 = 60$ for the rest, giving 180. Hence $P = \dfrac{180}{360} = \dfrac{1}{2}$.

DID YOU KNOW? **Counting and the National Lottery**

Combination counting explains why jackpots are so hard to win. In a lottery where you must match 5 numbers out of 49, the number of possible tickets is ${}^{49}C_5 = 1\,906\,884$: a single ticket has probability about 5.2×10^{-7} of winning. The same mathematics, run in reverse, is used by quality-control engineers in factories — including those bottling drinks in Douala — to decide how large a sample must be so that "at least one defective" is detected with high probability.

7 TOTAL PROBABILITY AND BAYES' THEOREM

In the previous section we used permutations and combinations to count favourable outcomes when all outcomes are equally likely. But many real problems are not about counting: they are about combining information that arrives in stages. A patient takes a test because she might be ill; an item is defective because it came from a particular machine; a student is late because he chose a particular means of transport. In all these situations there is a hidden **cause** (disease, machine, transport) and an observed **result** (positive test, defective item, lateness). This final

section gives the two master formulas for such problems: the **law of total probability**, which gathers all the ways a result can happen, and **Bayes' theorem**, which reverses the question: given the result, what is the probability of each cause?

7.1 Partition of the Sample Space

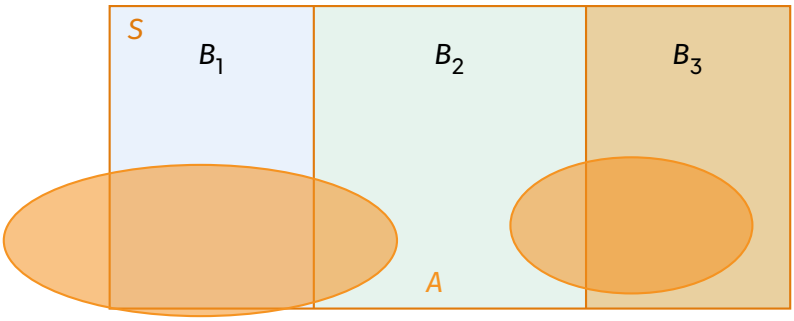
DEFINITION **Partition of a sample space**

Events $B_1, B_2, \dots, B_n$ form a **partition** of the sample space S if:

1. they are mutually exclusive: $B_i \cap B_j = \varnothing$ for $i \neq j$;
2. they are exhaustive: $B_1 \cup B_2 \cup \cdots \cup B_n = S$;
3. each has non-zero probability: $P(B_i) > 0$.

In particular $P(B_1) + P(B_2) + \cdots + P(B_n) = 1$.

Intuitively, a partition is a complete, non-overlapping list of "cases": exactly one of the B_i must occur. Typical examples: {disease, no disease}; {item from machine 1, machine 2, machine 3}; {travel by bus, by taxi, on foot}. The simplest partition of all is $\{B, B'\}$, an event and its complement — and indeed most examination questions use just two or three cases.



A partition B_1, B_2, B_3 of S. The event A crosses every strip: $A = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3)$, a disjoint union.

The key observation from the figure: whatever the event A, it can be split into disjoint pieces $A \cap B_1, A \cap B_2, \dots$, one piece inside each case. Since the pieces are mutually exclusive, their probabilities add.

7.2 The Law of Total Probability

PROPERTY **Law of total probability**

If $B_1, B_2, \dots, B_n$ form a partition of S, then for any event A,

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \cdots + P(A \mid B_n)P(B_n) = \sum_{i=1}^{n} P(A \mid B_i)\,P(B_i).$$

Proof (examinable and instructive). Since the B_i are exhaustive, $A = (A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_n)$. Since the B_i are mutually exclusive, so are the sets $A \cap B_i$, hence by the addition law $P(A) = \sum_i P(A \cap B_i)$. Now apply the multiplication law $P(A \cap B_i) = P(A \mid B_i)P(B_i)$ to each term. ■

On a two-stage tree (first the cause B_i, then the result A or A'), the law of total probability says simply: add the probabilities of all paths that end at A.

EXEMPLE **Lateness and transport**

A student at a school in Bamenda travels by bus with probability 0.5, by taxi with probability 0.3 and on foot with probability 0.2. The probabilities of arriving late are 0.1, 0.05 and 0.3 respectively. Then

$$P(\text{late}) = 0.1 \times 0.5 + 0.05 \times 0.3 + 0.3 \times 0.2 = 0.05 + 0.015 + 0.06 = 0.125.$$

The three products are the three paths leading to “late”; the sum is the total probability.

7.3 Bayes’ Theorem: Reversing the Conditioning

Now the deeper question. The student is late. What is the probability that he came on foot? The tree gives $P(\text{late} \mid \text{foot})$ directly, but we want $P(\text{foot} \mid \text{late})$ — the conditioning is reversed.

PROPERTY **Bayes’ theorem**

If $B_1, \dots, B_n$ form a partition of S and $P(A) > 0$, then for each i,

$$P(B_i \mid A) = \frac{P(A \mid B_i)\,P(B_i)}{\displaystyle\sum_{j=1}^{n} P(A \mid B_j)\,P(B_j)} = \frac{P(A \cap B_i)}{P(A)}.$$

Proof. By definition of conditional probability, $P(B_i \mid A) = \dfrac{P(A \cap B_i)}{P(A)}$. Write the numerator as $P(A \mid B_i)P(B_i)$ (multiplication law) and expand the denominator by the law of total probability. ■

The vocabulary is worth remembering: $P(B_i)$ is called the **prior** probability (before the evidence), and $P(B_i \mid A)$ the **posterior** probability (after observing A). Bayes’ theorem is the rule that updates the prior into the posterior.

WARNING $P(A \mid B)$ **is not** $P(B \mid A)$

"The probability of a positive test given the disease" and "the probability of the disease given a positive test" are different numbers, often dramatically different. Confusing them is the single most common error in this topic — and in real medical reasoning. Bayes' theorem exists precisely to pass from one to the other.

METHOD **The two-stage tree method for Bayes problems**

1. Draw the tree in the natural order: first the cause $B_1, \ldots, B_n$ (with their prior probabilities), then the result A or A' on each branch (conditional probabilities).
2. Compute every path probability by multiplying along the branches.
3. The denominator of Bayes' formula is the sum of all paths ending at A (total probability).
4. The answer is the ratio: (path through B_i ending at A) ÷ (sum of all paths ending at A).

This is exactly how the GCE examiner expects the solution to be presented: a labelled tree, then a clear ratio.

7.4 Application 1: Medical Testing

WORKED EXAMPLE 11 **A disease test**

A disease affects 1% of a population. A test detects the disease with probability 0.95 when it is present (**sensitivity**), and gives a false positive with probability 0.04 when the disease is absent. A person tests positive. What is the probability that he actually has the disease?

SOLUTION

Let D = "has the disease", T = "tests positive". Then $P(D) = 0.01$, $P(D') = 0.99$, $P(T \mid D) = 0.95$, $P(T \mid D') = 0.04$.
Step 1 — path probabilities.

$$P(T \cap D) = P(T \mid D)\,P(D) = 0.95 \times 0.01 = 0.0095,$$
$$P(T \cap D') = P(T \mid D')\,P(D') = 0.04 \times 0.99 = 0.0396.$$

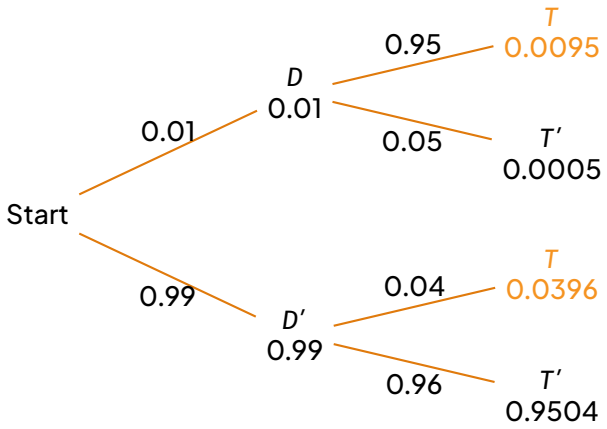
Step 2 — total probability of a positive test.

$$P(T) = 0.0095 + 0.0396 = 0.0491.$$

Step 3 — Bayes' ratio.

$$P(D \mid T) = \frac{P(T \cap D)}{P(T)} = \frac{0.0095}{0.0491} \approx 0.193.$$

Despite testing positive, the person has only about a 19% chance of having the disease!



Two-stage tree for the medical test. The two orange paths lead to a positive test T; Bayes' ratio is $\frac{0.0095}{0.0095 + 0.0396} \approx 0.193$.

WARNING **The denominator must contain ALL paths to A**

A frequent exam error is to write $P(D \mid T) = \frac{0.0095}{0.01}$ or $\frac{0.0095}{0.95}$. The denominator is $P(T)$, the sum of every path ending at T — including the false-positive path 0.0396. Forgetting the false positives is exactly the mistake that makes people overestimate the meaning of a positive test.

DID YOU KNOW? **Why rare diseases give surprising results**

The result $P(D \mid T) \approx 0.19$ shocks most people: the test is "95% accurate", yet a positive result means only about a 19% chance of illness. The reason is that the disease is rare. Out of 10 000 people, about 100 are ill (about 95 test positive), while 9 900 are healthy (about 396 false positives). So among the roughly 491 positive results, only about 95 are truly ill. When the prior probability $P(D)$ is tiny, even a small false-positive rate floods the positive group. This is why doctors confirm a positive screening test with a second, different test.

7.5 Application 2: Defective Items from Different Machines

WORKED EXAMPLE 12 **Two machines**

A factory in Douala produces 60% of its items on machine M_1 and 40% on machine M_2. Machine M_1 produces 2% defective items and M_2 produces 5% defective items. An item chosen at random is found to be defective. Find the probability that it came from machine M_2.

SOLUTION

Let F = "item is defective". Given: $P(M_1) = 0.6$, $P(M_2) = 0.4$, $P(F \mid M_1) = 0.02$, $P(F \mid M_2) = 0.05$.
Path probabilities to F:

$$P(F \cap M_1) = 0.02 \times 0.6 = 0.012, \qquad P(F \cap M_2) = 0.05 \times 0.4 = 0.020.$$

Total probability: $P(F) = 0.012 + 0.020 = 0.032$.
Bayes' ratio:

$$P(M_2 \mid F) = \frac{P(F \cap M_2)}{P(F)} = \frac{0.020}{0.032} = 0.625.$$

Although M_2 produces only 40% of the items, it accounts for 62.5% of the defective ones — because its defect rate is higher.

7.6 Application 3: Reversing the Transport Problem

Return to the Bamenda student: $P(\text{late}) = 0.125$, and the path through "on foot" has probability $0.3 \times 0.2 = 0.06$. Given that he is late,

$$P(\text{foot} \mid \text{late}) = \frac{0.06}{0.125} = 0.48.$$

Before knowing anything, the probability that he walked was only 0.2; the information "he is late" raises it to 0.48. Bayes' theorem is precisely the mathematics of updating a probability in the light of new evidence: the prior $P(B_i)$ becomes the posterior $P(B_i \mid A)$.

WORKED EXAMPLE 13 **Three causes — full Bayes with a three-branch partition**

Urn U_1 contains 3 red and 7 black balls, urn U_2 contains 6 red and 4 black balls, and urn U_3 contains 8 red and 2 black balls. An urn is chosen at random (each with probability $\frac{1}{3}$) and a ball is drawn. The ball is red. Find the probability that it came from urn U_3.

SOLUTION

Let R = "red ball". The conditional probabilities are

$$P(R \mid U_1) = \tfrac{3}{10} = 0.3, \qquad P(R \mid U_2) = \tfrac{6}{10} = 0.6, \qquad P(R \mid U_3) = \tfrac{8}{10} = 0.8.$$

Paths to R:

$$P(R \cap U_1) = 0.3 \times \tfrac{1}{3} = 0.1, \quad P(R \cap U_2) = 0.6 \times \tfrac{1}{3} = 0.2, \quad P(R \cap U_3) = 0.8 \times \tfrac{1}{3} = \tfrac{0.8}{3}.$$

Total probability:

$$P(R) = \tfrac{1}{3}(0.3 + 0.6 + 0.8) = \tfrac{1.7}{3}.$$

Bayes' ratio:

$$P(U_3 \mid R) = \frac{0.8/3}{1.7/3} = \frac{0.8}{1.7} = \frac{8}{17} \approx 0.471.$$

The prior probability of U_3 was $\tfrac{1}{3} \approx 0.333$; observing a red ball raises it to $\tfrac{8}{17} \approx 0.471$, because U_3 is the urn richest in red balls.

7.7 Exam Technique and Consolidation

METHOD **GCE strategy for Bayes questions**

1. Define events clearly in words (D, T, M_1, …) and list every given probability with correct conditioning.
2. Draw the labelled two-stage tree and write the path probabilities at the tips.
3. For "find $P(A)$", add the relevant paths (total probability).
4. For "given A, find $P(B_i \mid A)$", write the ratio path/sum of paths to A, showing both numbers explicitly.
5. Give the final answer as a fraction or a decimal to 3 significant figures, and add one sentence of interpretation.

EXERCISE 8

A bag is chosen from three bags B_1, B_2, B_3 with probabilities $0.5, 0.3, 0.2$. The bags contain red balls in proportions $0.4, 0.5, 0.75$ respectively. A ball is drawn and is red. Find the probability that it came from bag B_3.

SOLUTION **Exercise 1**

Let R = "red ball". Paths to R: $0.4 \times 0.5 = 0.20$; $0.5 \times 0.3 = 0.15$; $0.75 \times 0.2 = 0.15$. Total: $P(R) = 0.20 + 0.15 + 0.15 = 0.50$. Hence

$$P(B_3 \mid R) = \frac{0.15}{0.50} = 0.3.$$

EXERCISE 9

In a school, 55% of the candidates study science and 45% study arts. The probability of passing the GCE is 0.8 for a science candidate and 0.7 for an arts candidate. (a) Find the probability that a randomly chosen candidate passes. (b) A candidate passed; find the probability that she is a science candidate.

SOLUTION **Exercise 2**

Let S = "science", A = "arts", P = "passes". (a) Total probability:

$$P(P) = 0.8 \times 0.55 + 0.7 \times 0.45 = 0.44 + 0.315 = 0.755.$$

(b) Bayes' ratio:

$$P(S \mid P) = \frac{0.44}{0.755} \approx 0.583.$$

KEY POINTS **Key points of the section**

- A **partition** splits S into mutually exclusive, exhaustive cases $B_1, \dots, B_n$.
- **Total probability**: $P(A) = \sum_i P(A \mid B_i)P(B_i)$ — add all paths leading to A.
- **Bayes' theorem**: $P(B_i \mid A) = \dfrac{P(A \mid B_i)P(B_i)}{\sum_j P(A \mid B_j)P(B_j)}$ — the ratio of one path to all paths through A.
- $P(A \mid B) \neq P(B \mid A)$; Bayes reverses the conditioning, updating the **prior** into the **posterior**.
- With a rare cause, even a very accurate test gives mostly false positives: always compare the true-positive path with all positive paths.

8 INTEGRATION ACTIVITY

9 ACTIVITIES OF INTEGRATION, ASSESSMENT AND REMEDIATION

9.1 Aims of the activity

This integration activity mobilises, in one authentic investigation, all the skills developed in this chapter:

- identifying a random experiment, its outcomes and its sample space;
- distinguishing events and computing **classical probabilities** using **combinations**;
- constructing and reading a **tree diagram** with conditional probabilities;
- applying the **multiplication law**, the **law of total probability** and **Bayes' theorem**;
- using the **complement rule** for "at least one" problems;
- distinguishing **selection with replacement** from **selection without replacement**;
- communicating a probabilistic conclusion in clear, non-technical language for a decision maker.

By the end of the activity, each group should be able to convert a real-life security problem into a probability model, compute the relevant probabilities rigorously, and defend a practical recommendation before the class.

9.2 Situation

The Mobile Money Fraud Investigation.

Madam Neba runs a busy mobile money kiosk at Commercial Avenue in Bamenda. Every day she processes hundreds of deposits and withdrawals. Recently, her operator's fraud-monitoring unit sent her a worrying report. After a careful audit of transaction records across her region, the unit established the following facts, which we shall take as exact for this investigation:

- fraudulent withdrawal attempts occur on 2% of all transactions;
- the operator's automatic security system flags (marks as suspicious) 95% of all fraudulent transactions;
- unfortunately, the same system also wrongly flags 3% of all genuine transactions.

Whenever a transaction is flagged, Madam Neba must decide what to do. Blocking a genuine customer's transaction causes anger and loss of business; letting a fraudulent transaction through causes direct financial loss. She asks your class, as trainee data analysts, to answer the following questions.

1. If a transaction is flagged by the system, what is the probability that it is actually fraudulent? Madam Neba guesses "about 95%, since the system catches 95% of frauds". Is she right?
2. On a typical morning, the system flags 5 transactions. Assuming these flagged transactions behave independently, what is the probability that at least one of them is actually fraudulent?
3. As a final check, an auditor examines a batch of 20 transactions known to contain exactly 2 fraudulent ones. The auditor selects 4 transactions at random (without replacement) for a detailed manual review. What is the probability that the review detects at least one of the two frauds?

Finally, each group must write a short **advisory note** (at most ten lines) to Madam Neba recommending whether flagged transactions should be blocked automatically or verified manually first, justifying the recommendation with the computed probabilities.

9.3 Tasks

Work in groups of three or four. Show all reasoning; a numerical answer without justification earns little credit.

1. **Modelling.** Define the experiment and the two events F = "the transaction is fraudulent" and S = "the transaction is flagged by the system". Write down, in probability notation, the three numerical data of the situation. What is the sample space if we classify one transaction according to both criteria?
2. **Tree diagram.** Draw a two-stage tree diagram (first stage: fraudulent or genuine; second stage: flagged or not flagged). Label every branch with its probability and compute the probability of each of the four terminal outcomes.
3. **Total probability.** Using the tree, compute $P(S)$, the probability that a transaction is flagged.
4. **Bayes' theorem.** Compute $P(F \mid S)$, the probability that a flagged transaction is actually fraudulent. Compare with Madam Neba's guess of 95% and explain, in two or three sentences, why the true value is so much lower even though the system catches 95% of frauds.
5. **Several flagged transactions.** Assuming independence, compute the probability that at least one of 5 flagged transactions is fraudulent. (Hint: use the complement rule.)
6. **Counting techniques.** For the auditor's sample of 4 transactions chosen from 20 containing 2 frauds, compute:

   (a) the total number of possible samples;

   (b) the number of samples containing no fraud;

   (c) the probability that the sample contains at least one fraud.

   Explain clearly why combinations (and not permutations) are appropriate here, and why this is selection without replacement.
7. **Advisory note.** Write your recommendation to Madam Neba. It must mention the value of $P(F \mid S)$, the consequence of blocking genuine customers, and a concrete proposal (for example: manual verification, a second confirmation step, or a phone call to the customer).

8. **Plenary.** Each group presents its tree, its value of $P(F \mid S)$ and its recommendation in three minutes. Compare the recommendations across groups.

WARNING **The two classic confusions**

- $P(S \mid F) = 0.95$ is not the same as $P(F \mid S)$. The first is given; the second is what Bayes' theorem computes. Confusing them is the most frequent error in this type of problem.
- The auditor samples without replacement: use combinations, not powers. Writing $\left(\frac{18}{20}\right)^4$ would only be correct if each transaction were replaced before the next draw.

9.4 Model answer

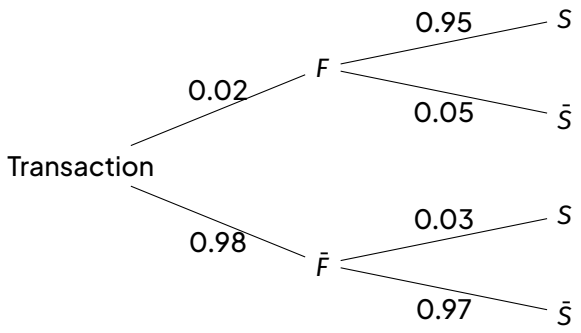
1. Modelling. The experiment is: "observe one transaction and record whether it is fraudulent and whether it is flagged". Let F = "the transaction is fraudulent" and S = "the transaction is flagged". The data are

$$P(F) = 0.02, \qquad P(S \mid F) = 0.95, \qquad P(S \mid \bar{F}) = 0.03,$$

where $\bar{F}$ = "the transaction is genuine", with $P(\bar{F}) = 1 - 0.02 = 0.98$. The sample space, classifying by both criteria, is

$$\Omega = \{F \cap S,\ F \cap \bar{S},\ \bar{F} \cap S,\ \bar{F} \cap \bar{S}\}.$$

2. Tree diagram.



Two-stage tree for the fraud investigation

The four terminal probabilities (multiplication law) are:

$$P(F \cap S) = 0.02 \times 0.95 = 0.019,$$
$$P(F \cap \bar{S}) = 0.02 \times 0.05 = 0.001,$$
$$P(\bar{F} \cap S) = 0.98 \times 0.03 = 0.0294,$$
$$P(\bar{F} \cap \bar{S}) = 0.98 \times 0.97 = 0.9506.$$

Check: $0.019 + 0.001 + 0.0294 + 0.9506 = 1$. ✓

3. Total probability. A transaction is flagged either because it is fraudulent and caught, or genuine and wrongly flagged:

$$P(S) = P(F \cap S) + P(\bar{F} \cap S) = 0.019 + 0.0294 = 0.0484.$$

So about 4.84% of all transactions are flagged.

4. Bayes' theorem.

$$P(F \mid S) = \frac{P(F \cap S)}{P(S)} = \frac{P(S \mid F)\,P(F)}{P(S)} = \frac{0.95 \times 0.02}{0.0484} = \frac{0.019}{0.0484} \approx 0.3926.$$

Only about 39.3% of flagged transactions are actually fraudulent — far below Madam Neba's guess of 95%. The reason is that fraud is rare (2%): the genuine transactions are so numerous (98%) that even a small false-alarm rate of 3% produces 0.0294 worth of false alarms, which outnumber the 0.019 of true alarms. In other words, most flagged transactions are innocent.

5. At least one fraud among 5 flagged transactions. For a flagged transaction, $P(\bar{F} \mid S) = 1 - 0.3926 = 0.6074$. Assuming independence,

$$P(\text{at least one fraud}) = 1 - (0.6074)^5 \approx 1 - 0.0827 \approx 0.9173.$$

There is about a 91.7% chance that at least one of the five flagged transactions is genuinely fraudulent — so the flags must be taken seriously even though each individual flag is more likely than not to be a false alarm.

6. Auditor's sample (without replacement).

1. Total number of samples of 4 from 20:

$$\binom{20}{4} = \frac{20 \times 19 \times 18 \times 17}{4 \times 3 \times 2 \times 1} = 4845.$$

2. Samples containing no fraud: choose all 4 from the 18 genuine ones:

$$\binom{18}{4} = \frac{18 \times 17 \times 16 \times 15}{24} = 3060.$$

3. By the complement rule:

$$P(\text{at least one fraud}) = 1 - \frac{3060}{4845} = \frac{1785}{4845} \approx 0.3684.$$

Combinations are appropriate because the order in which the auditor picks the transactions is irrelevant — only the set selected matters. The draws are without replacement since a transaction cannot be reviewed twice. (With replacement, the answer would have been $1 - (0.9)^4 = 0.3439$, a different value.)

7. Advisory note (example). "Dear Madam Neba, our analysis shows that only about 39% of flagged transactions are actually fraudulent: more than 6 out of 10 flags are false alarms on honest customers. We therefore advise you **not** to block flagged transactions automatically. Instead, verify each flag manually (call the customer, check identity) before blocking. This is worthwhile because among any 5 flagged transactions there is about a 92% chance that at least one is a real fraud."

WARNING **Remediation targets**

If your group wrote $P(F \mid S) = 0.95$, revisit Task 4: you confused a conditional probability with its converse. If in Task 6 you used powers of $\frac{18}{20}$, revisit the with/without replacement distinction: sampling several items from a finite batch at once is always without replacement.

10 METHODS AND WORKED EXAMPLES

10.1 Skill 1: Computing a Classical (Theoretical) Probability

METHOD **Calculating $P(A) = \dfrac{n(A)}{n(S)}$**

1. Identify the **experiment** and check that all outcomes are **equally likely** (fair coin, fair die, well-shuffled cards, random draw).
2. List or count the **sample space** S; write $n(S)$.
3. List or count the outcomes favourable to the event A; write $n(A)$.
4. Compute $P(A) = \dfrac{n(A)}{n(S)}$ and simplify the fraction. Give the answer as a fraction, decimal or percentage as requested.
5. Check: the answer must satisfy $0 \le P(A) \le 1$. If not, recount.

Reflexes: a standard die has 6 faces; a pack of cards has 52 cards (4 suits, 13 ranks); when two dice are thrown $n(S) = 36$ and a table of sums avoids omissions.

WARNING **Equally likely outcomes**

The formula $P(A) = \dfrac{n(A)}{n(S)}$ is valid only when all outcomes of the sample space are equally likely. For two dice, the 36 ordered pairs are equally likely, but the 11 possible sums are not: a sum of 2 occurs once while a sum of 7 occurs six times. Never apply the formula to a list of non-equiprobable outcomes.

WORKED EXAMPLE 14 **Two dice at a school fair**

At a fund-raising game in a college in Bamenda, two fair dice are thrown once. Find the probability that (a) the sum of the scores is 9; (b) the sum is at least 10; (c) the two dice show the same number.

SOLUTION

The experiment is throwing two fair dice; all 36 ordered pairs are equally likely, so $n(S) = 6 \times 6 = 36$.

(a) Sum equal to 9. The favourable outcomes are

$$(3,6),\ (4,5),\ (5,4),\ (6,3), \quad \text{so } n(A) = 4.$$

Hence $P(\text{sum} = 9) = \dfrac{4}{36} = \dfrac{1}{9}$.
(b) Sum at least 10. This means sum = 10, 11 or 12:

$$\begin{aligned} \text{sum } 10 &: (4,6), (5,5), (6,4) \quad (3 \text{ outcomes}) \\ \text{sum } 11 &: (5,6), (6,5) \quad (2 \text{ outcomes}) \\ \text{sum } 12 &: (6,6) \quad (1 \text{ outcome}) \end{aligned}$$

So $n(B) = 3 + 2 + 1 = 6$ and $P(B) = \dfrac{6}{36} = \dfrac{1}{6}$.
(c) Same number on both dice. The doubles are $(1,1), (2,2), \ldots, (6,6)$, giving $n(C) = 6$, so

$$P(C) = \frac{6}{36} = \frac{1}{6}.$$

All three answers lie between 0 and 1, as required.

10.2 Skill 2: Estimating an Empirical Probability

METHOD **Using relative frequency**

1. Identify the total number of **trials** N (throws, observations, sampled items).
2. Count the number f of trials in which the event occurred.
3. Compute the **relative frequency** $P(A) \approx \dfrac{f}{N}$.
4. Interpret: the larger N is, the more reliable the estimate (law of large numbers).
5. Use the estimate to predict: expected number of occurrences in n future trials $\approx n \times \dfrac{f}{N}$.

Reflex: empirical probability comes from data; theoretical probability comes from reasoning about equally likely outcomes.

WORKED EXAMPLE 15 **Defective loaves in a bakery**

A bakery in Douala inspected 500 loaves and found that 35 were underweight. (a) Estimate the probability that a randomly chosen loaf is underweight. (b) On a day when 2 000 loaves are produced, how many underweight loaves should be expected? (c) Is this a the-

oretical or an empirical probability? Justify.

SOLUTION

(a) The number of trials is $N = 500$ and the frequency of "underweight" is $f = 35$. Hence

$$P(\text{underweight}) \approx \frac{f}{N} = \frac{35}{500} = 0.07.$$

(b) Expected number $\approx n \times 0.07 = 2\,000 \times 0.07 = 140$ loaves.
(c) This is an **empirical** (experimental) probability: it is obtained from observed data, not from a symmetry argument. The loaves are not "equally likely outcomes" of a known sample space; the estimate 0.07 could change if a larger sample were taken, and it becomes more reliable as N grows.

10.3 Skill 3: Applying the Addition Laws

METHOD **Mutually exclusive or not?**

1. Define the events A and B clearly in words.
2. Ask: can A and B happen at the same time? If no, they are **mutually exclusive** and $P(A \cap B) = 0$.
3. If mutually exclusive: $P(A \cup B) = P(A) + P(B)$.
4. If not: use the general law $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
5. For "neither", use the complement: $P(\text{neither}) = 1 - P(A \cup B)$.

Reflex: never add probabilities blindly; subtract the overlap or you count it twice.

WORKED EXAMPLE 16 **Drawing a card from a pack**

A card is drawn at random from a standard pack of 52 cards. Find the probability that it is (a) a king or a queen; (b) a heart or a king; (c) neither a heart nor a king.

SOLUTION

Each card is equally likely, so probabilities are counted out of 52.
(a) Let K = "king", Q = "queen". A card cannot be both, so the events are mutually exclusive:

$$P(K \cup Q) = P(K) + P(Q) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}.$$

(b) Let H = "heart". A card can be both a heart and a king (the king of hearts), so the events are not mutually exclusive. We have $P(H) = \frac{13}{52}$, $P(K) = \frac{4}{52}$ and $P(H \cap K) = \frac{1}{52}$. By the general addition law:

$$P(H \cup K) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}.$$

(c) "Neither a heart nor a king" is the complement of $H \cup K$:

$$P(\text{neither}) = 1 - P(H \cup K) = 1 - \frac{4}{13} = \frac{9}{13}.$$

Check: there are 13 hearts plus 3 non-heart kings = 16 "bad" cards, leaving 52 - 16 = 36 good cards, and $\frac{36}{52} = \frac{9}{13}$. ✓

10.4 Skill 4: Multiplication Law, Conditional Probability and Independence

METHOD **Using $P(A \cap B) = P(A)\,P(B \mid A)$**

1. Translate the question: "probability of B given that A has occurred" means $P(B \mid A) = \frac{P(A \cap B)}{P(A)}$, provided $P(A) \neq 0$.
2. For successive draws, update the sample space after each draw (the composition of the bag/pack changes if there is no replacement).
3. Multiply along the chain: $P(A \cap B) = P(A) \times P(B \mid A)$.
4. To test **independence**, check whether $P(B \mid A) = P(B)$, equivalently $P(A \cap B) = P(A)P(B)$.
5. For **exhaustive** events, check that the probabilities sum to 1.

Reflex: "given that" ⇒ conditional probability; "and" ⇒ intersection (multiply); "or" ⇒ union (add carefully).

WARNING **Mutually exclusive is not independent**

Do not confuse the two notions. **Mutually exclusive** means $A \cap B = \varnothing$ (they cannot occur together). **Independent** means $P(A \cap B) = P(A)P(B)$ (knowing one occurred does not change the probability of the other). Two non-impossible mutually exclusive events are never independent, since $P(A \cap B) = 0 \neq P(A)P(B)$.

WORKED EXAMPLE 17 **Students studying French and Computer Science**

In an Upper Sixth class of 40 students, 25 study French, 18 study Computer Science and 10 study both. A student is chosen at random. (a) Find the probability that the student studies Computer Science, given that the student studies French. (b) Are the events "studies French" and "studies Computer Science" independent? (c) Show whether these two events are exhaustive.

SOLUTION

Let F = "studies French" and C = "studies Computer Science". Then

$$P(F) = \frac{25}{40} = \frac{5}{8}, \qquad P(C) = \frac{18}{40} = \frac{9}{20}, \qquad P(F \cap C) = \frac{10}{40} = \frac{1}{4}.$$

(a) By the definition of conditional probability,

$$P(C \mid F) = \frac{P(C \cap F)}{P(F)} = \frac{\frac{1}{4}}{\frac{5}{8}} = \frac{1}{4} \times \frac{8}{5} = \frac{2}{5} = 0.4.$$

Interpretation: among the 25 French students, 10 also study Computer Science, and $\frac{10}{25} = \frac{2}{5}$. ✓

(b) Test independence: $P(C \mid F) = 0.4$ while $P(C) = \frac{9}{20} = 0.45$. Since $P(C \mid F) \neq P(C)$, the events are not independent. (Equivalently, $P(F)P(C) = \frac{5}{8} \times \frac{9}{20} = \frac{9}{32} \neq \frac{1}{4} = P(F \cap C)$.)

(c) Exhaustive events must cover the whole sample space, i.e. their union must have probability 1:

$$P(F \cup C) = \frac{5}{8} + \frac{9}{20} - \frac{1}{4} = \frac{25 + 18 - 10}{40} = \frac{33}{40} \neq 1.$$

Since 7 students study neither subject, F and C are not exhaustive.

10.5 Skill 5: Tree Diagrams and Selection With or Without Replacement

METHOD **Solving with a tree diagram**

1. Identify the **stages** of the experiment (first draw, second draw, …).
2. Draw branches for all possible outcomes at each stage and write the probability on each branch. Probabilities leaving any node must sum to 1.
3. With replacement: branch probabilities stay the same at every stage. Without replacement: reduce both the number of favourable items and the total by 1 after each draw.
4. Multiply probabilities along a path to get the probability of that path.

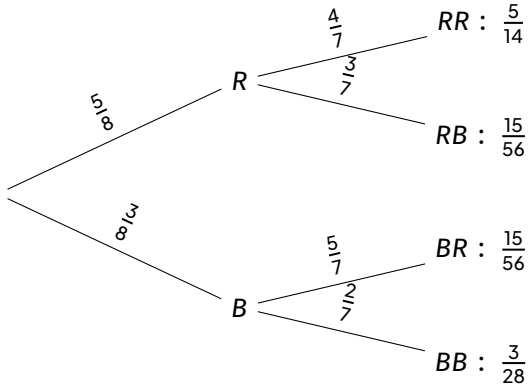
5. Add the probabilities of all paths corresponding to the required event.
6. Check: the probabilities of all complete paths must sum to 1.

WORKED EXAMPLE 18 **Balls drawn from a bag**

A bag contains 5 red and 3 blue balls. Two balls are drawn at random, one after the other, without replacement. (a) Draw a tree diagram for the experiment. (b) Find the probability that both balls are red. (c) Find the probability that the two balls are of different colours. (d) Repeat (b) if the first ball is replaced before the second draw.

SOLUTION

(a) At the first draw: $P(R) = \frac{5}{8}$, $P(B) = \frac{3}{8}$. Since there is no replacement, after a red first draw the bag has 4 red and 3 blue out of 7; after a blue first draw it has 5 red and 2 blue out of 7.



Tree diagram for two draws without replacement; path probabilities are obtained by multiplying along the branches.

(b) Path R then R: multiply along the branches:

$$P(RR) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}.$$

(c) Different colours means RB or BR:

$$P(RB) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56},$$
$$P(BR) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56},$$
$$P(\text{different}) = \frac{15}{56} + \frac{15}{56} = \frac{30}{56} = \frac{15}{28}.$$

Check of the whole tree: $\frac{20}{56} + \frac{15}{56} + \frac{15}{56} + \frac{6}{56} = \frac{56}{56} = 1$. ✓

(d) With replacement, the bag is unchanged for the second draw, so the draws are independent:

$$P(RR) = \frac{5}{8} \times \frac{5}{8} = \frac{25}{64}.$$

Note that $\frac{25}{64} \approx 0.391 > \frac{5}{14} \approx 0.357$: replacement makes a repeated colour slightly more likely here.

10.6 Skill 6: Using Permutations and Combinations in Probability

METHOD **Counting with ${}^nC_r$ and ${}^nP_r$**

1. Decide whether **order matters**. Committee, hand of cards, selection ⇒ order does not matter: use ${}^nC_r = \frac{n!}{r!(n-r)!}$. Arrangement, ranking, podium ⇒ order matters: use ${}^nP_r = \frac{n!}{(n-r)!}$.
2. Count $n(S)$, the total number of possible selections/arrangements, using the same rule as for $n(A)$.
3. Count $n(A)$ by splitting the population into categories and multiplying the choices for each category.
4. Compute $P(A) = \frac{n(A)}{n(S)}$ and simplify.
5. For "at least one" questions, it is often faster to use the complement: $P(\text{at least one}) = 1 - P(\text{none})$.

Reflex: numerator and denominator must count the same kind of object (both unordered or both ordered).

WORKED EXAMPLE 19 **Choosing a delegation**

A school in Yaoundé must send a delegation of 4 students chosen at random from 7 boys and 5 girls. Find the probability that the delegation contains (a) exactly 2 girls; (b) at least one girl; (c) all four students of the same sex.

SOLUTION

There are 12 students and the delegation is an unordered selection, so

$$n(S) = {}^{12}C_4 = \frac{12!}{4!\,8!} = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} = 495.$$

(a) Exactly 2 girls means 2 girls from 5 and 2 boys from 7:

$$n(A) = {}^5C_2 \times {}^7C_2 = 10 \times 21 = 210, \qquad P(A) = \frac{210}{495} = \frac{14}{33}.$$

(b) At least one girl. Use the complement: "no girl" means 4 boys from 7:

$$P(\text{no girl}) = \frac{{}^7C_4}{{}^{12}C_4} = \frac{35}{495} = \frac{7}{99},$$

$$P(\text{at least one girl}) = 1 - \frac{7}{99} = \frac{92}{99}.$$

(c) All of the same sex means 4 girls or 4 boys (mutually exclusive):

$$n = {}^5C_4 + {}^7C_4 = 5 + 35 = 40, \qquad P = \frac{40}{495} = \frac{8}{99}.$$

Each answer lies in $[0, 1]$; the direct counting in (c) confirms the complement figure used in (b).

10.7 Skill 7: Total Probability and Bayes' Theorem

METHOD **Applying Bayes' theorem**

1. Identify the **partition** of the sample space: events $B_1, B_2, \ldots, B_n$ that are mutually exclusive and exhaustive (e.g. which box, which factory, which route).
2. Identify the event A that is observed (e.g. "defective item", "positive test").
3. Write down $P(B_i)$ and the conditional probabilities $P(A \mid B_i)$.
4. Compute the **total probability**: $P(A) = \sum_i P(B_i)\,P(A \mid B_i)$.
5. Reverse the conditioning with **Bayes' theorem**:

$$P(B_k \mid A) = \frac{P(B_k)\,P(A \mid B_k)}{\sum_i P(B_i)\,P(A \mid B_i)}.$$

6. A tree diagram with the B_i at the first stage makes the whole computation visual and safe.

WORKED EXAMPLE 20 **Two factories supplying a market**

A market in Buea is supplied with bottled groundnut oil by two factories X and Y. Factory

X supplies 60% of the bottles and Y supplies 40%. It is known that 2% of the bottles from X and 5% of those from Y are defective. A bottle is bought at random. (a) Find the probability that it is defective. (b) Given that the bottle is defective, find the probability that it came from factory Y.

SOLUTION

Let D = "bottle is defective". The events X and Y form a partition: $P(X) = 0.6$, $P(Y) = 0.4$, with $P(D \mid X) = 0.02$ and $P(D \mid Y) = 0.05$.
(a) By the law of total probability,

$$\begin{aligned} P(D) &= P(X)P(D \mid X) + P(Y)P(D \mid Y) \\ &= (0.6)(0.02) + (0.4)(0.05) \\ &= 0.012 + 0.020 = 0.032. \end{aligned}$$

So 3.2% of all bottles on the market are defective.
(b) By Bayes' theorem,

$$P(Y \mid D) = \frac{P(Y)\,P(D \mid Y)}{P(D)} = \frac{0.020}{0.032} = \frac{20}{32} = \frac{5}{8} = 0.625.$$

Interpretation. Although factory Y supplies only 40% of the bottles, it produces $\frac{5}{8} = 62.5\%$ of the defective ones, because its defect rate is higher. Note that $P(Y \mid D) \neq P(D \mid Y)$: confusing these two conditional probabilities is the classic Bayes trap. Check: $P(X \mid D) = \frac{0.012}{0.032} = 0.375$, and $0.375 + 0.625 = 1$. ✓

11 EXERCISES

11.1 I check my knowledge

EXERCISE 10 **Multiple choice questions**

For each question, choose the single correct answer.

1. A fair die is thrown once. The probability of obtaining a prime number is:

   (a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{2}{3}$

2. Two events A and B are mutually exclusive, with $P(A) = 0.3$ and $P(B) = 0.5$. Then $P(A \cup B)$ equals:

   (a) 0.15 (b) 0.8 (c) 0.65 (d) 0.2

3. If A and B are independent events with $P(A) = 0.4$ and $P(B) = 0.25$, then $P(A \cap B)$ equals:

   (a) 0.65 (b) 0.1 (c) 0.15 (d) 0.55

4. A card is drawn from a pack of 52. The probability that it is a king or a heart is:

   (a) $\frac{17}{52}$ (b) $\frac{15}{52}$ (c) $\frac{4}{13}$ (d) $\frac{1}{13}$

EXERCISE 11 **True or false**

State whether each statement is true or false, justifying your answer with a short argument or a counter-example.

1. If two events are mutually exclusive, then they are independent.
2. For any event A, $P(A) + P(A') = 1$.
3. If $P(A \mid B) = P(A)$, then A and B are independent.
4. The empirical probability of an event always equals its theoretical probability.

EXERCISE 12 **Definitions**

Define each of the following terms precisely, giving one example of each:

1. a random experiment and its sample space;
2. mutually exclusive events;

3. exhaustive events;
4. independent events;
5. the conditional probability $P(A \mid B)$.

EXERCISE 13 **Direct calculations**

1. A letter is chosen at random from the word CAMEROON. Find the probability that it is a vowel.
2. Two fair coins are tossed. Write down the sample space and find the probability of obtaining exactly one head.
3. Given $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, find $P(A \cup B)$ and $P(A \mid B)$.
4. A fair coin is tossed 200 times and lands heads 118 times. State the empirical probability of heads, compare it with the theoretical probability, and explain what you expect as the number of tosses increases.

11.2 I practise

EXERCISE 14 **Balls of the same colour**

A bag contains 6 red and 4 black balls. Two balls are drawn at random. Find the probability that they are of the same colour:

1. when the first ball is replaced before the second draw;
2. when the first ball is not replaced.

Compare the two answers and comment.

EXERCISE 15 **Physics and Chemistry students**

In a class of 40 students, 18 study Physics, 15 study Chemistry and 7 study both subjects. A student is chosen at random from the class.

1. Illustrate the situation with a Venn diagram.
2. Find the probability that the student studies Physics or Chemistry.
3. Find the probability that the student studies neither subject.

4. Given that the chosen student studies Physics, find the probability that he or she also studies Chemistry.

EXERCISE 16 **Committee selection**

A committee of 4 people is to be chosen at random from 5 men and 6 women. Using combinations, find the probability that the committee contains:

1. exactly 2 women;
2. at least one man.

EXERCISE 17 **Rainy weekend in Buea**

The probability that it rains on any day in Buea in July is 0.6, independently of other days. Using a tree diagram for a 3-day weekend, find the probability that:

1. it rains on exactly one day;
2. it rains on at least one day.

11.3 I prepare for the GCE

EXERCISE 18 **Two dice and independence**

Two fair dice are thrown and the scores are added.

1. List the outcomes for which the sum is greater than 8, and show that $P(\text{sum} > 8) = \frac{5}{18}$. (3 marks)
2. Find the probability that at least one die shows a 6. (2 marks)
3. Given that the sum is greater than 8, find the probability that at least one die shows a 6. (3 marks)
4. Are the events “sum greater than 8” and “at least one 6” independent? Justify your answer carefully. (2 marks)

EXERCISE 19 **Defective items and Bayes’ theorem**

Three machines A, B and C produce 50%, 30% and 20% respectively of a factory’s output.

The percentages of defective items produced by A, B and C are 2%, 4% and 5% respectively.

1. Draw a fully labelled tree diagram representing this situation. (3 marks)
2. Show that the probability that a randomly chosen item is defective is 0.032. (3 marks)
3. An item is chosen at random and found to be defective. Use Bayes' theorem to find the probability that it came from machine A. (3 marks)
4. From which machine is a defective item most likely to have come? Justify. (2 marks)

EXERCISE 20 **Drawing cards without replacement**

Cards are drawn one by one, at random and without replacement, from a full pack of 52 playing cards.

1. Find the probability that the first two cards drawn are both aces. (3 marks)
2. Find the probability that the first heart appears on the third draw. (4 marks)
3. Find the probability that the second card drawn is a heart, given that the first card drawn was a heart. (2 marks)
4. Using combinations, find the probability that a hand of 5 cards contains exactly 3 hearts. (3 marks)

12 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple choice questions**

1. The prime numbers on a fair die are 2, 3, 5. There are 3 favourable outcomes out of 6 equally likely outcomes. Hence $P(\text{prime}) = \frac{3}{6} = \frac{1}{2}$. The correct answer is (c).
2. For mutually exclusive events, $P(A \cup B) = P(A) + P(B) = 0.3 + 0.5 = 0.8$. The correct answer is (b).
3. For independent events, $P(A \cap B) = P(A) \times P(B) = 0.4 \times 0.25 = 0.1$. The correct answer is (b).
4. Let K be the event "king" and H be the event "heart". There are 4 kings and 13 hearts, but the king of hearts is counted twice. So $P(K \cup H) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$. The correct answer is (c).

SOLUTION **Exercise 2 — True or false**

1. False. Mutually exclusive events cannot occur simultaneously, so $P(A \cap B) = 0$. For independence we would need $P(A \cap B) = P(A)P(B)$. If $P(A) > 0$ and $P(B) > 0$, then $P(A)P(B) > 0$, so they cannot be independent. Counter-example: rolling a die, $A = \{1, 2\}$, $B = \{3, 4\}$ are mutually exclusive but not independent.
2. True. By definition of the complement, A and A' partition the sample space, so $P(A) + P(A') = 1$.
3. True. $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$. If $P(A \mid B) = P(A)$, then $P(A \cap B) = P(A)P(B)$, which is exactly the definition of independence (provided $P(B) > 0$).
4. False. Empirical probability is based on observed frequencies and varies from experiment to experiment. It approaches the theoretical probability as the number of trials increases (Law of Large Numbers), but it is not always equal.

SOLUTION **Exercise 3 — Definitions**

1. Random experiment: An experiment whose outcome cannot be predicted with certainty before it is performed, but whose set of all possible outcomes is known. Sample space (S): The set of all possible outcomes of a random experiment. Example: Tossing a fair coin. The experiment is random; the sample space is $S = \{H, T\}$.

2. Mutually exclusive events: Two events A and B are mutually exclusive (or disjoint) if they cannot occur at the same time, i.e. $A \cap B = \emptyset$. Example: In a single die roll, $A = \{\text{even}\}$ and $B = \{\text{odd}\}$ are mutually exclusive.

3. Exhaustive events: A set of events $\{E_1, E_2, \ldots, E_n\}$ is exhaustive if their union is the whole sample space, i.e. $E_1 \cup E_2 \cup \cdots \cup E_n = S$. Example: In a die roll, $E_1 = \{1, 2, 3\}$, $E_2 = \{4, 5, 6\}$ are exhaustive.

4. Independent events: Two events A and B are independent if the occurrence of one does not affect the probability of the other, i.e. $P(A \cap B) = P(A)P(B)$. Example: Tossing two fair coins. Let $A =$ "first coin is heads", $B =$ "second coin is heads". Then $P(A \cap B) = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2} = P(A)P(B)$.

5. Conditional probability $P(A \mid B)$: The probability of event A occurring given that event B has already occurred, defined as $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ for $P(B) > 0$. Example: Draw a card from a standard deck. $A =$ "card is a king", $B =$ "card is a face card". $P(A \mid B) = \frac{4/52}{12/52} = \frac{1}{3}$.

SOLUTION **Exercise 4 — Direct calculations**

1. The word CAMEROON has 8 letters: C, A, M, E, R, O, O, N. Vowels: A, E, O, O (4 vowels). $P(\text{vowel}) = \frac{4}{8} = \frac{1}{2}$.

2. Sample space for two fair coins: $S = \{HH, HT, TH, TT\}$. Exactly one head: $\{HT, TH\}$. $P(\text{exactly one head}) = \frac{2}{4} = \frac{1}{2}$.

3. $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.2 = 0.9$. $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5} = 0.4$.

4. Empirical probability of heads $= \frac{118}{200} = 0.59$. Theoretical probability $= 0.5$. As the number of tosses increases, the empirical probability tends to approach the theoretical probability (Law of Large Numbers).

SOLUTION **Exercise 5 — Balls of the same colour**

Bag: 6 red (R), 4 black (B). Total = 10.

1. With replacement: Draws are independent. $P(\text{both red}) = \frac{6}{10} \times \frac{6}{10} = \frac{36}{100} = 0.36$. $P(\text{both black}) = \frac{4}{10} \times \frac{4}{10} = \frac{16}{100} = 0.16$. $P(\text{same colour}) = 0.36 + 0.16 = 0.52 = \frac{13}{25}$.

2. Without replacement: Draws are dependent. $P(\text{both red}) = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$. $P(\text{both black}) = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$. $P(\text{same colour}) = \frac{1}{3} + \frac{2}{15} = \frac{5}{15} + \frac{2}{15} = \frac{7}{15} \approx 0.4667$.

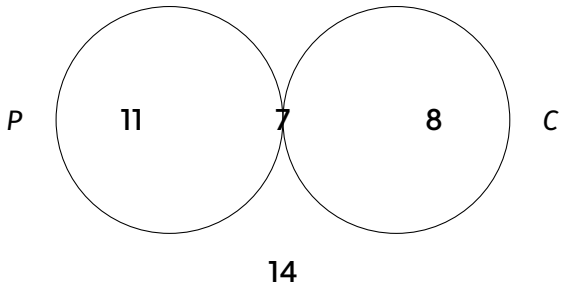
Comment: The probability is higher with replacement (0.52) than without (≈ 0.467). With replacement, the composition of the bag remains the same for the second draw, keeping

the probability of drawing a red ball high. Without replacement, drawing a red ball first reduces the number of red balls left, making a second red less likely; similarly for black. The effect of "without replacement" reduces the chance of two balls of the same colour because the first draw changes the proportions.

SOLUTION **Exercise 6 — Physics and Chemistry students**

Let P = studies Physics, C = studies Chemistry. Total students = 40. $n(P) = 18$, $n(C) = 15$, $n(P \cap C) = 7$.

1. Venn diagram: Two overlapping circles. Region $P \cap C$: 7. Region P only: $18 - 7 = 11$. Region C only: $15 - 7 = 8$. Outside both: $40 - (11 + 7 + 8) = 14$.



Venn diagram for Physics and Chemistry students

2. $P(P \cup C) = \frac{n(P \cup C)}{40} = \frac{11+7+8}{40} = \frac{26}{40} = \frac{13}{20} = 0.65$.
3. $P(\text{neither}) = \frac{14}{40} = \frac{7}{20} = 0.35$.
4. $P(C \mid P) = \frac{P(C \cap P)}{P(P)} = \frac{7/40}{18/40} = \frac{7}{18}$.

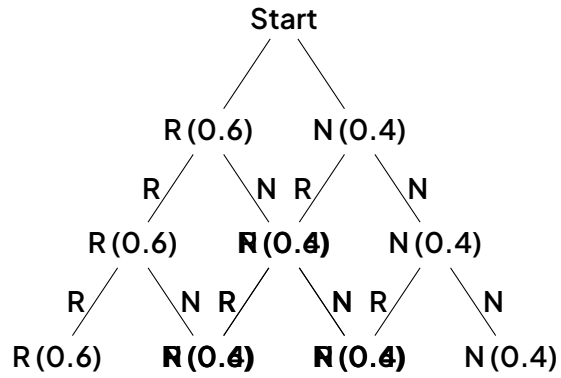
SOLUTION **Exercise 7 — Committee selection**

Total people = 11 (5 men, 6 women). Committee of 4 chosen at random. Total number of possible committees: $\binom{11}{4} = 330$.

1. Exactly 2 women ⇒ 2 women and 2 men. Number of ways = $\binom{6}{2} \times \binom{5}{2} = 15 \times 10 = 150$. $P(\text{exactly 2 women}) = \frac{150}{330} = \frac{5}{11}$.
2. At least one man = complement of "no men" (i.e., all 4 women). Number of all-women committees = $\binom{6}{4} = 15$. $P(\text{at least one man}) = 1 - \frac{15}{330} = 1 - \frac{1}{22} = \frac{21}{22}$.

SOLUTION **Exercise 8 — Rainy weekend in Buea**

P(rain on a day) = 0.6, P(no rain) = 0.4. Three independent days.



Tree diagram for 3-day weekend (R = rain, N = no rain)

1. Exactly one day of rain: sequences RNN, NRN, NNR. Each has probability 0.6 × 0.4 × 0.4 = 0.096. Total P = 3 × 0.096 = 0.288.
2. At least one day of rain = complement of no rain at all (NNN). P(NNN) = 0.4^3 = 0.064. P(at least one rain) = 1 - 0.064 = 0.936.

SOLUTION **Exercise 9 — Two dice and independence**

Two fair dice: sample space of 36 equally likely ordered pairs (i, j).

1. Sum > 8 means sum = 9, 10, 11, 12. Outcomes: Sum 9: (3,6), (4,5), (5,4), (6,3) → 4 outcomes. Sum 10: (4,6), (5,5), (6,4) → 3 outcomes. Sum 11: (5,6), (6,5) → 2 outcomes. Sum 12: (6,6) → 1 outcome. Total favourable = 4+3+2+1 = 10. $P(\text{sum} > 8) = \frac{10}{36} = \frac{5}{18}$. (3 marks)
2. At least one die shows a 6. Complement: no 6 on either die. Each die has 5 non-6 faces → 5 × 5 = 25 outcomes. $P(\text{at least one 6}) = 1 - \frac{25}{36} = \frac{11}{36}$. (2 marks)
3. Given sum > 8, we restrict to the 10 outcomes listed above. Among these, which have at least one 6? Sum 9: (3,6), (6,3) → 2 outcomes. Sum 10: (4,6), (6,4) → 2 outcomes. Sum 11: (5,6), (6,5) → 2 outcomes. Sum 12: (6,6) → 1 outcome. Total = 7 outcomes. $P(\text{at least one 6} \mid \text{sum} > 8) = \frac{7}{10} = 0.7$. (3 marks)
4. Check independence: $P(\text{sum} > 8) = \frac{5}{18} \approx 0.2778$, $P(\text{at least one 6}) = \frac{11}{36} \approx 0.3056$. Product = $\frac{5}{18} \times \frac{11}{36} = \frac{55}{648} \approx 0.0849$. $P(\text{sum} > 8 \cap \text{at least one 6}) = \frac{7}{36} \approx 0.1944$. Since $\frac{7}{36} \neq \frac{55}{648}$, the events are not independent. (2 marks)

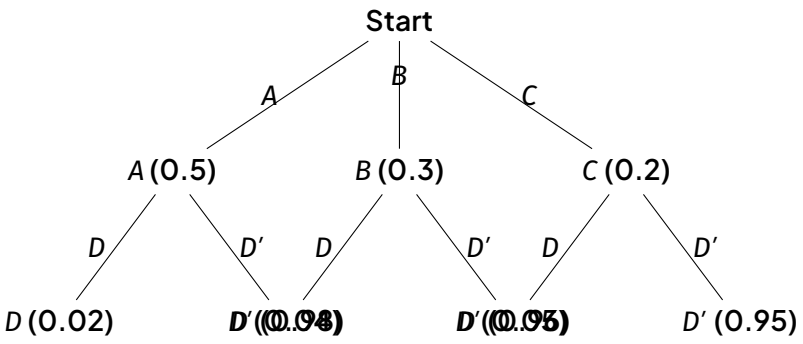
Mark scheme / Examiner expectations:

- Part (i): List outcomes clearly (3 marks: 1 for listing, 1 for count, 1 for probability).
- Part (ii): Use complement or direct count (2 marks).
- Part (iii): Correct conditional probability using reduced sample space (3 marks).
- Part (iv): State definition of independence, compute both sides, conclude not equal (2 marks).

SOLUTION **Exercise 10 — Defective items and Bayes' theorem**

Machines: A (50%), B (30%), C (20%). Defective rates: $P(D|A) = 0.02$, $P(D|B) = 0.04$, $P(D|C) = 0.05$.

1. Tree diagram:



Tree diagram for machine output and defectives

(3 marks)

2. Total probability of defective: $P(D) = P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C) = 0.5 \times 0.02 + 0.3 \times 0.04 + 0.2 \times 0.05 = 0.01 + 0.012 + 0.01 = 0.032$. (3 marks)

3. Bayes' theorem for $P(A|D)$: $P(A|D) = \frac{P(A)P(D|A)}{P(D)} = \frac{0.5 \times 0.02}{0.032} = \frac{0.01}{0.032} = \frac{10}{32} = \frac{5}{16} = 0.3125$. (3 marks)

4. Compare $P(A|D) = 0.3125$, $P(B|D) = \frac{0.3 \times 0.04}{0.032} = \frac{0.012}{0.032} = 0.375$, $P(C|D) = \frac{0.2 \times 0.05}{0.032} = \frac{0.01}{0.032} = 0.3125$. Machine B has the highest posterior probability (0.375). So a defective item is most likely to have come from machine B. (2 marks)

Mark scheme / Examiner expectations:

- Tree diagram: branches labelled with probabilities, all outcomes shown (3 marks).
- Total probability: correct formula and arithmetic (3 marks).
- Bayes' theorem: correct substitution and simplification (3 marks).

- Comparison of posterior probabilities, clear conclusion (2 marks).

SOLUTION **Exercise 11 — Drawing cards without replacement**

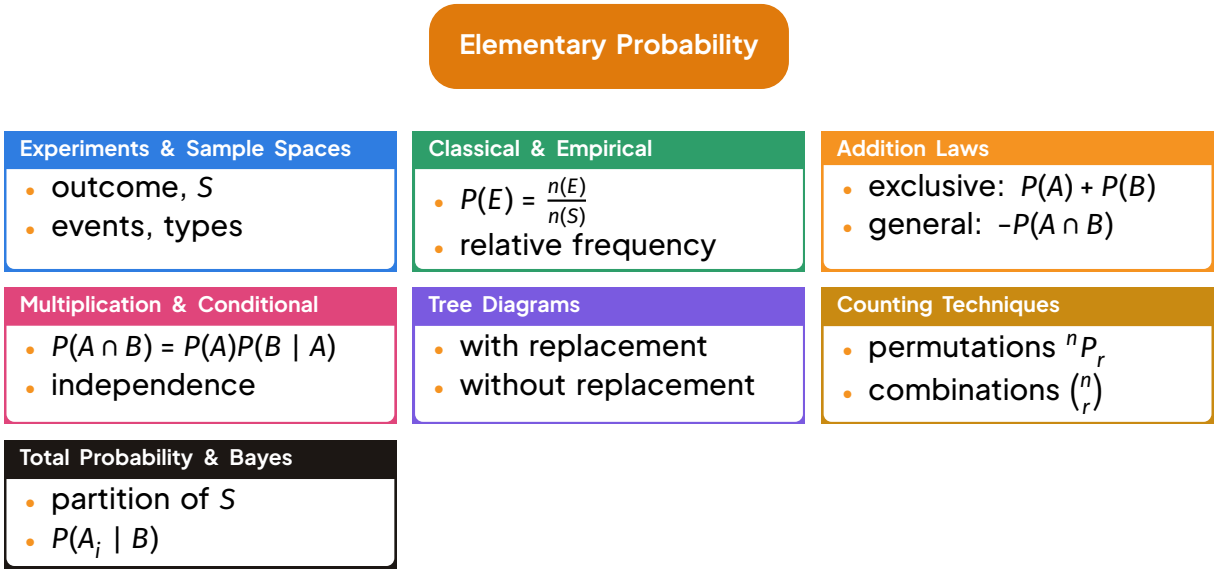
Standard deck: 52 cards, 4 aces, 13 hearts.

1. First two cards both aces. $P(\text{1st ace}) = \frac{4}{52}$, $P(\text{2nd ace} \mid \text{1st ace}) = \frac{3}{51}$. $P(\text{both aces}) = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$. (3 marks)
2. First heart appears on the third draw. This means: 1st not heart, 2nd not heart, 3rd heart. $P(\text{1st not heart}) = \frac{39}{52} = \frac{3}{4}$. $P(\text{2nd not heart} \mid \text{1st not heart}) = \frac{38}{51}$. $P(\text{3rd heart} \mid \text{first two not hearts}) = \frac{13}{50}$. Product: $\frac{39}{52} \times \frac{38}{51} \times \frac{13}{50} = \frac{39\times38\times13}{52\times51\times50}$. Simplify: $\frac{39}{52} = \frac{3}{4}$, $\frac{38}{51} = \frac{38}{51}$, $\frac{13}{50} = \frac{13}{50}$. $= \frac{3\times38\times13}{4\times51\times50} = \frac{1482}{10200} = \frac{247}{1700} \approx 0.1453$. (4 marks)
3. $P(\text{2nd heart} \mid \text{1st heart})$. After drawing one heart, 51 cards remain, 12 hearts. $P = \frac{12}{51} = \frac{4}{17}$. (2 marks)
4. Hand of 5 cards, exactly 3 hearts. Number of ways: choose 3 hearts from 13, and 2 non-hearts from 39. $\binom{13}{3}\times\binom{39}{2} = 286\times741 = 211926$. Total 5-card hands: $\binom{52}{5} = 2598960$.
   $P = \frac{211926}{2598960} = \frac{\binom{13}{3}\binom{39}{2}}{\binom{52}{5}} \approx 0.0815$. (3 marks)

Mark scheme / Examiner expectations:

- (i) Correct conditional probability multiplication (3 marks).
- (ii) Sequence of three dependent events, correct fractions (4 marks).
- (iii) Simple conditional probability after one heart removed (2 marks).
- (iv) Correct use of combinations, numerator and denominator (3 marks).

13 SUMMARY SHEET



Mind map of the chapter Elementary Probability

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions.

- A **random experiment** is a process whose outcome cannot be predicted with certainty; an **outcome** is a single possible result; the **sample space** S is the set of all outcomes.
- An **event** is a subset of S. Events may be simple, compound, sure (S), impossible ($\varnothing$), complementary (A'), mutually exclusive ($A \cap B = \varnothing$), or exhaustive (their union is S).
- **Classical probability**: for equally likely outcomes, $P(E) = \dfrac{n(E)}{n(S)}$.
- **Empirical probability**: $P(E) \approx \dfrac{\text{frequency of } E}{\text{number of trials}}$ for a large number of trials.
- **Conditional probability**: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, $P(B) > 0$.
- A and B are **independent** when $P(A \cap B) = P(A)\,P(B)$, equivalently $P(A \mid B) = P(A)$.

Laws and formulas to know.

- $0 \le P(E) \le 1$; $P(S) = 1$; $P(\varnothing) = 0$; $P(A') = 1 - P(A)$.
- Addition law (mutually exclusive): $P(A \cup B) = P(A) + P(B)$.

- Addition law (general): $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Multiplication law: $P(A \cap B) = P(A)\,P(B \mid A) = P(B)\,P(A \mid B)$; for independent events $P(A \cap B) = P(A)P(B)$.
- Total probability: if $A_1, \ldots, A_n$ partition S, then $P(B) = \sum_i P(A_i)\,P(B \mid A_i)$.
- Bayes' theorem: $P(A_i \mid B) = \dfrac{P(A_i)\,P(B \mid A_i)}{\sum_j P(A_j)\,P(B \mid A_j)}$.
- Counting: ${}^nP_r = \dfrac{n!}{(n-r)!}$ (order matters); $\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$ (order does not matter); $P(E) = \dfrac{\text{favourable selections}}{\text{total selections}}$.

Methods.

- Always start by writing the sample space (or its size) and checking that outcomes are equally likely.
- For sequential experiments, draw a **tree diagram**: multiply along branches, add between branches ending in the required event.
- With replacement: probabilities stay constant along the tree (independent stages). Without replacement: denominators decrease by one at each stage.
- "At least one" problems: use the complement, $P(\text{at least one}) = 1 - P(\text{none})$.
- For selection problems, decide first whether order matters (permutation) or not (combination).

Common mistakes.

- Adding $P(A) + P(B)$ without checking that the events are mutually exclusive.
- Confusing independence ($P(A \cap B) = P(A)P(B)$) with mutual exclusivity ($P(A \cap B) = 0$): exclusive events with positive probabilities are never independent.
- Forgetting to reduce the denominator in selection without replacement.
- Inverting the condition: $P(A \mid B) \neq P(B \mid A)$ in general.
- Probabilities on a tree branch not summing to 1 at each node.

GCE tips.

- Give final answers as simplified fractions, or decimals to at least 3 s.f. when required.
- In Bayes-type questions, define events clearly in words before computing; examiners award marks for correct notation.
- Show the tree diagram: it earns method marks even if the final arithmetic slips.
- Check your answer lies in $[0, 1]$; a probability greater than 1 signals an error.

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